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# ==== run.py ====
# -*- coding: utf-8 -*-
"""系统启动入口。
运行：
    python run.py                启动并自动打开浏览器
    python run.py --no-browser   只启动服务，不打开浏览器
    python run.py --dev          开发模式（开启代码热重载与调试器）
    python run.py --port 5001    指定端口
默认以 127.0.0.1:5000 启动，就绪后自动打开浏览器。
"""
import os
import sys
import threading
import time
import webbrowser
if hasattr(sys.stdout, "reconfigure"):
    sys.stdout.reconfigure(encoding="utf-8")   # 控制台中文正常显示
from app import create_app
from app.models import db
PORT = 5000
for i, a in enumerate(sys.argv):
    if a == "--port" and i + 1 < len(sys.argv):
        PORT = int(sys.argv[i + 1])
URL = "http://127.0.0.1:%d" % PORT
NO_BROWSER = "--no-browser" in sys.argv
DEV = "--dev" in sys.argv
app = create_app()
def open_browser_later(delay=1.2):
    """等服务工作起来再打开浏览器，避免打开时页面还没就绪。
    放在后台线程里，不用 sleep 阻塞服务启动；线程设为守护线程，
    用户 Ctrl+C 停止服务时不会残留。
    """
    def run():
        time.sleep(delay)
        try:
            webbrowser.open(URL)
        except Exception:
            pass            # 打开失败不影响服务本身
    t = threading.Thread(target=run, daemon=True)
    t.start()
if __name__ == "__main__":
    with app.app_context():
        db.create_all()     # 首次运行时自动建表
        from app.utils.schema import ensure_schema
        added = ensure_schema(db)     # 旧库补齐新增字段
        if added:
            print("已补充字段：%s" % "、".join(added))
    print("=" * 58)
    print("   历史遗留废弃矿山自然恢复适宜性评价系统  Ｖ１．０")
    print("=" * 58)
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    print("   访问地址：%s" % URL)
    print("   演示账号：admin      密码：admin123")
    print("   停止运行：按 Ctrl+C，或直接关闭本窗口")
    print("=" * 58)
    if not NO_BROWSER and not DEV:
        print("   正在打开浏览器……")
    print()
    # 开发模式才启用热重载（重载会二次启动进程，故此时不自动开浏览器）
    if not NO_BROWSER and not DEV:
        open_browser_later()
    app.run(host="127.0.0.1", port=PORT, debug=DEV,
    use_reloader=DEV)
# ==== app/__init__.py ====
# -*- coding: utf-8 -*-
"""Flask 应用工厂。"""
from datetime import date
from flask import Flask
from flask_login import LoginManager
from flask_wtf.csrf import CSRFProtect, CSRFError
from app.config import Config
from app.models import db, User
from app.utils.constants import ROLE_LABELS
login_manager = LoginManager()
login_manager.login_view = "auth.login"
login_manager.login_message = "请先登录"
csrf = CSRFProtect()
@login_manager.user_loader
def load_user(user_id):
    return db.session.get(User, int(user_id))
def create_app(config_class=Config):
    app = Flask(__name__)
    app.config.from_object(config_class)
    db.init_app(app)
    login_manager.init_app(app)
    # CSRF 防护：WTF_CSRF_ENABLED 只对 FlaskForm 生效，页面上那些手写 POST
    表单
    # （账号增删改、项目 / 图斑删除等）以前是没校验的；同时模板里的 csrf_token()
    # 也需要它才会存在——少了这一行，保存过一次 AHP 权重后 /ahp/ 直接 500。
    csrf.init_app(app)
    # 模板里展示时间统一走 localdt：库里存 UTC，展示成本地时间
    from app.utils.timefmt import to_local
    app.jinja_env.filters["localdt"] = to_local
    # 注册视图（各模块路由）
    from app.views import auth, project, unit, standard,
    constraint, \
        ahp, indicator, evaluate, report, search
    app.register_blueprint(auth.bp)
    app.register_blueprint(project.bp)
    app.register_blueprint(unit.bp)
    app.register_blueprint(standard.bp)
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app.register_blueprint(constraint.bp)
app.register_blueprint(ahp.bp)
app.register_blueprint(indicator.bp)
app.register_blueprint(evaluate.bp)
app.register_blueprint(report.bp)
app.register_blueprint(search.bp)
# 全局登录保护：除登录页与静态资源外，均需登录
from flask import request, redirect, url_for,
send_from_directory
from flask_login import current_user, login_user
PUBLIC_ENDPOINTS = {"auth.login", "auth.register",
"static", "favicon"}
# 浏览器默认会请求根路径下的 /favicon.ico。
# 页面里虽已用 <link rel="icon"> 指向 static 目录，但出错页等场景仍会
# 回退到这个路径，补一条路由免得日志里一直刷 404。
@app.route("/favicon.ico")
def favicon():
    return send_from_directory(app.static_folder,
"favicon.ico",
                               mimetype="image/vnd.micro
soft.icon")
@app.before_request
def require_login():
    # 截图模式（仅用于生成说明书配图）：自动以管理员身份登录
    if app.config.get("SCREENSHOT_MODE"):
        if (request.endpoint not in ("auth.login",
"auth.register", "static")
                and not current_user.is_authenticated):
            admin = User.query.filter_by(role="admin").f
irst()
            if admin:
                login_user(admin)
        return None
    if request.endpoint in PUBLIC_ENDPOINTS or
request.endpoint is None:
        return None
    if not current_user.is_authenticated:
        return redirect(url_for("auth.login"))
    return None
# 侧边栏导航：按论文评价流程的顺序编排，分为「评价流程」与「系统管理」两组。
# 不使用模块编号，避免出现与实际流程不一致的代号。
# 每项带图标名（内联 SVG，见 templates/_macros.html 的 ICONS）
# 与可选的待办角标（该步骤还没做完的图斑数）。
FLOW_LINKS = [
    ("项目与矿山档案", "/project/", "folder"),
    ("政策与标准知识库", "/standard/", "book"),
    ("限制性指标判定", "/constraint/", "shield"),
    ("微地质单元", "/unit/", "layers"),
    ("指标体系与权重", "/ahp/", "sliders"),
    ("指标数据采集", "/indicator/", "database"),
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        ("适宜性定量评价", "/evaluate/", "target"),
        ("报告与结果输出", "/report/", "file"),
    ]
    #  「用户与操作日志」已并入「个人设置」，不再是独立菜单，
    #  因此系统管理组只留一个入口。
    ADMIN_LINKS = [
        ("个人设置", "/profile", "user"),
    ]
    @app.context_processor
    def inject_nav():
        path = request.path or "/"
        def hit(prefix):
            """当前路径是否属于某个入口（/unit/ 也匹配 /unit/3）。"""
            return path.startswith(prefix.rstrip("/"))
        badges = {}
        if current_user.is_authenticated:
            from app.services import dashboard as dash
            badges = dash.nav_badges()
        def build(links):
            return [{"label": label, "href": href, "icon":
    icon,
                     "active": hit(href), "badge":
    badges.get(href, 0)}
                    for label, href, icon in links]
        groups = [
            {"title": "评价流程", "links": build(FLOW_LINKS)},
            {"title": "系统管理", "links": build(ADMIN_LINKS)},
        ]
        #  面包屑用的当前页名：按路径命中的入口取，未命中时给个通用名
        if path == "/":
            page_title = "工作台"
        elif path.startswith("/profile"):
            page_title = "个人设置"
        elif path.startswith(("/users", "/logs")):
            page_title = "账号与日志"
        elif path.startswith("/search"):
            page_title = "搜索结果"
        else:
            page_title = "工作台"
            for label, href, _icon in FLOW_LINKS:
                if hit(href):
                    page_title = label
                    break
        #  层级引导用：项目是归档单位，图斑才是评价对象。
        #  下游各页面按图斑组织数据，因此需要提示「哪些项目下还没有图斑」。
        from app.models import Project, MinePatch
        projects = Project.query.order_by(Project.id).all()
        return {
            "nav_groups": groups,
            "home_active": path == "/",
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                "page_title": page_title,
                "today": date.today().strftime("%Y-%m-%d"),
                "search_q": request.args.get("q", ""),
                "role_labels": ROLE_LABELS,
                "project_count": len(projects),
                "patch_count": MinePatch.query.count(),
                "projects_without_patch": [p for p in projects
                                           if p.patches.count()
    == 0],
            }
    # 操作审计：统一在响应阶段记录成功的写操作（操作日志）
    from app.services import audit
    @app.after_request
    def audit_writes(response):
        if request.method in ("POST", "PUT", "PATCH",
    "DELETE"):
            audit.record(request.endpoint,
    response.status_code)
        return response
    # 错误页：给出可读提示，而不是默认的英文错误页
    from flask import render_template
    @app.errorhandler(403)
    def forbidden(e):
        return render_template("error.html", code=403,
                               message="没有权限执行该操作"), 403
    @app.errorhandler(404)
    def not_found(e):
        return render_template("error.html", code=404,
                               message="请求的页面不存在"), 404
    @app.errorhandler(CSRFError)
    def csrf_error(e):
        """令牌过期或来源不符时给一句人话，并把人送回上一页。
        原来校验一失败就是默认英文 400 页，或者（对手写表单）干脆什么都不发生，
        用户只会觉得「点了没反应」。回跳目标只取本站地址，避免开放重定向。
        """
        from flask import flash, redirect, url_for
        flash("页面停留时间过久或提交来源校验未通过，请重新提交一次", "error")
        referrer = request.referrer or ""
        if referrer.startswith(request.host_url):
            return redirect(referrer)
        return redirect(url_for("auth.index"))
    return app
# ==== app/config.py ====
# -*- coding: utf-8 -*-
"""应用配置。"""
import os
from datetime import timedelta
BASE_DIR = os.path.abspath(os.path.dirname(__file__))
class Config:
    SECRET_KEY = os.environ.get("SECRET_KEY",
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    "mine-restoration-dev-key")
    # 单机部署，SQLite 数据库文件放在应用目录下
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE_DIR, "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    # 禁止跨站请求伪造（CSRF），登录与表单提交依赖此项
    WTF_CSRF_ENABLED = True
    # 模板改动立即生效，不必重启服务。
    # Flask 在非 debug 下默认把模板缓存到进程里，改完模板刷新浏览器还是旧页面，
    # 容易被误当成「改了没生效」。本系统只在本机运行，这点开销可以忽略。
    TEMPLATES_AUTO_RELOAD = True
    # 「记住我」：登录时勾选则下发持久 cookie，关闭浏览器后仍可自动登录
    REMEMBER_COOKIE_DURATION = timedelta(days=30)
    REMEMBER_COOKIE_HTTPONLY = True
    REMEMBER_COOKIE_SAMESITE = "Lax"
    # 影像与数据文件上传（遥感影像挂接、批量导入）
    UPLOAD_FOLDER = os.path.join(BASE_DIR, "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024          # 单次上传上限
    32 MB
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
    ALLOWED_DATA_EXT = {"csv", "xlsx", "xls"}
    # 头像只用浏览器能直接显示的格式（不含 bmp / tif），并单独限制体积
    ALLOWED_AVATAR_EXT = {"png", "jpg", "jpeg", "gif",
    "webp"}
    MAX_AVATAR_LENGTH = 4 * 1024 * 1024            # 4 MB
# ==== app/models.py ====
# -*- coding: utf-8 -*-
"""数据模型：系统全部数据表。
本文件由甲在阶段 0 一次性写完，作为两人独立开发的地基。
约定：两人只读此文件；确需改表时，由乙提需求、甲统一修改。
"""
import json
from datetime import datetime
from flask_login import UserMixin
from flask_sqlalchemy import SQLAlchemy
db = SQLAlchemy()
class User(UserMixin, db.Model):
    """用户（账户与权限）。"""
    __tablename__ = "users"
    id = db.Column(db.Integer, primary_key=True)
    username = db.Column(db.String(64), unique=True,
    nullable=False, index=True)
    password_hash = db.Column(db.String(256),
    nullable=False)
    # 职务只有两种：管理员 admin（可管理账号与他人的职务） / 操作员 operator
    role = db.Column(db.String(20), nullable=False,
    default="operator")
    active = db.Column(db.Boolean, nullable=False,
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    default=True)                                                    301
    created_at = db.Column(db.DateTime,                              302
    default=datetime.utcnow)                                         303
    # ---- 个人资料（用户可在「个人设置」里自行维护）----                304
    display_name = db.Column(db.String(64), nullable=True)           305
    # 昵称                                                           306
    email = db.Column(db.String(128), nullable=True)                 307
    phone = db.Column(db.String(32), nullable=True)                  308
    gender = db.Column(db.String(8), nullable=True)                  309
    # 男 / 女 / 保密                                                 310
    avatar = db.Column(db.String(256), nullable=True)                311
    # 头像，相对 static 的路径                                       312
    @property                                                        313
    def is_admin(self):                                              314
        return self.role == "admin"                                  315
    @property                                                        316
    def shown_name(self):                                            317
        """界面上显示的名字：优先用昵称，没填就用账号名。"""          318
        return self.display_name or self.username                    319
    @property                                                        320
    def initial(self):                                               321
        """没有头像时用首字占位。"""                                  322
        return (self.shown_name or "?")[0].upper()                   323
    def to_dict(self):                                               324
        return {"id": self.id, "username": self.username,            325
                "display_name": self.display_name,                   326
    "gender": self.gender,                                           327
                "role": self.role, "active": self.active}            328
class OperationLog(db.Model):                                        329
    """操作日志。"""                                                 330
    __tablename__ = "operation_logs"                                 331
    id = db.Column(db.Integer, primary_key=True)                     332
    user_id = db.Column(db.Integer, db.ForeignKey("users.id"         333
    ), nullable=True)                                                334
    action = db.Column(db.String(128), nullable=False)               335
    # 操作类型                                                       336
    target_type = db.Column(db.String(64), nullable=True)            337
    # 对象类型                                                       338
    target_id = db.Column(db.Integer, nullable=True)                 339
    # 对象主键                                                       340
    detail = db.Column(db.Text, nullable=True)                       341
    # 操作详情                                                       342
    created_at = db.Column(db.DateTime,                              343
    default=datetime.utcnow, index=True)                             344
class Project(db.Model):                                             345
    """评价项目（项目与矿山档案）。"""                               346
    __tablename__ = "projects"                                       347
    id = db.Column(db.Integer, primary_key=True)                     348
    name = db.Column(db.String(128), nullable=False)                 349
    code = db.Column(db.String(64), nullable=True)                   350
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    # 项目编号
    description = db.Column(db.Text, nullable=True)
    status = db.Column(db.String(20), default="进行中")
    # 进行中 / 已完成 / 已归档
    created_by = db.Column(db.Integer,
    db.ForeignKey("users.id"), nullable=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    patches = db.relationship("MinePatch",
    backref="project", lazy="dynamic",
                              cascade="all, delete-orphan")
class MinePatch(db.Model):
    """矿山图斑。"""
    __tablename__ = "mine_patches"
    id = db.Column(db.Integer, primary_key=True)
    project_id = db.Column(db.Integer,
    db.ForeignKey("projects.id"), nullable=False)
    name = db.Column(db.String(128), nullable=False)
    admin_region = db.Column(db.String(128), nullable=True)
    # 所在行政区
    center_x = db.Column(db.Float, nullable=True)
    # 中心经度 / 横坐标
    center_y = db.Column(db.Float, nullable=True)
    # 中心纬度 / 纵坐标
    area_ha = db.Column(db.Float, nullable=True)
    # 图斑面积（公顷）
    mineral_type = db.Column(db.String(64), nullable=True)
    # 开采矿种
    mining_method = db.Column(db.String(64), nullable=True)
    # 开采方式
    stop_years = db.Column(db.Float, nullable=True)
    # 停产年限
    responsible_entity_missing = db.Column(db.Boolean,
    default=True)  # 责任主体是否缺失
    # 遥感底图 / 无人机正射影像挂接
    image_path = db.Column(db.String(256), nullable=True)
    # 相对 static 的路径
    image_note = db.Column(db.String(256), nullable=True)
    # 影像说明（如来源、时相）
    image_uploaded_at = db.Column(db.DateTime,
    nullable=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    units = db.relationship("MicroUnit", backref="patch",
    lazy="dynamic",
                            cascade="all, delete-orphan")
class ReferenceEcosystem(db.Model):
    """参照生态系统：标准化基准值，与图斑绑定。"""
    __tablename__ = "reference_ecosystems"
    id = db.Column(db.Integer, primary_key=True)
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    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    coverage_benchmark = db.Column(db.Float, default=75.0)
    # 植被覆盖度基准 %
    shannon_benchmark = db.Column(db.Float, default=0.7)
    # Shannon-Wiener 基准
    dominant_species = db.Column(db.String(256),
    nullable=True)  # 建群种
    soil_benchmark = db.Column(db.String(256),
    nullable=True)    # 土壤基准
class Standard(db.Model):
    """地方标准 / 政策文件（政策与标准知识库）。"""
    __tablename__ = "standards"
    id = db.Column(db.Integer, primary_key=True)
    province = db.Column(db.String(32), nullable=False)
    # 省份
    name = db.Column(db.String(256), nullable=False)
    # 名称
    doc_number = db.Column(db.String(128), nullable=True)
    # 标准编号 / 文号
    publish_date = db.Column(db.String(32), nullable=True)
    implement_date = db.Column(db.String(32), nullable=True)
    # 类别：地方标准 / 政策文件
    category = db.Column(db.String(32), nullable=False,
    default="地方标准")
    summary = db.Column(db.Text, nullable=True)
    # 特点概述
    rules = db.relationship("StandardRule",
    backref="standard",
                            cascade="all, delete-orphan")
class StandardRule(db.Model):
    """结构化规则：将标准条文拆解为可执行规则。"""
    __tablename__ = "standard_rules"
    id = db.Column(db.Integer, primary_key=True)
    standard_id = db.Column(db.Integer,
    db.ForeignKey("standards.id"), nullable=False)
    item = db.Column(db.String(128), nullable=False)
    # 判定项
    operator = db.Column(db.String(8), nullable=False)
    # 比较符：>  >=  <  <=  ==
    threshold = db.Column(db.String(64), nullable=True)
    # 阈值
    condition_type = db.Column(db.String(32),
    default="必要条件")   # 必要条件 / 充分条件
    region = db.Column(db.String(64), nullable=True)
    # 适用分区（宁夏分区差异化）
    note = db.Column(db.Text, nullable=True)
class ConstraintCheck(db.Model):
    """限制性判定记录。"""
    __tablename__ = "constraint_checks"
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    id = db.Column(db.Integer, primary_key=True)                    451
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat        452
    ches.id"), nullable=False)                                      453
    policy_pass = db.Column(db.Boolean, nullable=True)              454
    # 政策约束                                                      455
    location_pass = db.Column(db.Boolean, nullable=True)            456
    # 区位约束                                                      457
    pollution_pass = db.Column(db.Boolean, nullable=True)           458
    # 土壤污染                                                      459
    geo_pass = db.Column(db.Boolean, nullable=True)                 460
    # 地质灾害                                                      461
    farmland_pass = db.Column(db.Boolean, nullable=True)            462
    # 耕地属性                                                      463
    passed = db.Column(db.Boolean, nullable=True)                   464
    # 总体是否通过                                                  465
    reason = db.Column(db.Text, nullable=True)                      466
    # 不通过原因                                                    467
    checked_at = db.Column(db.DateTime,                             468
    default=datetime.utcnow)                                        469
class MicroUnit(db.Model):                                          470
    """微地质单元：岩质边坡 / 碎石边坡 / 平台。"""                  471
    __tablename__ = "micro_units"                                   472
    id = db.Column(db.Integer, primary_key=True)                    473
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat        474
    ches.id"), nullable=False)                                      475
    unit_type = db.Column(db.String(32), nullable=False)            476
    # 单元类型                                                      477
    area = db.Column(db.Float, nullable=True)                       478
    # 面积                                                          479
    ratio = db.Column(db.Float, nullable=True)                      480
    # 面积占比（自动计算）                                          481
    description = db.Column(db.Text, nullable=True)                 482
    # 现场描述                                                      483
class Indicator(db.Model):                                          484
    """指标定义：指标树，含角色与启用单元类型。"""                  485
    __tablename__ = "indicators"                                    486
    id = db.Column(db.Integer, primary_key=True)                    487
    code = db.Column(db.String(32), unique=True,                    488
    nullable=False)  # 指标编码                                    489
    name = db.Column(db.String(64), nullable=False)                 490
    # 指标名称                                                      491
    level = db.Column(db.Integer, nullable=False)                   492
    # 1/2/3 级                                                      493
    parent_code = db.Column(db.String(32), nullable=True)           494
    # 父指标编码                                                    495
    # 角色：土壤污染指标同时作为限制性指标与评价指标               496
    is_constraint = db.Column(db.Boolean, default=False)            497
    is_evaluative = db.Column(db.Boolean, default=True)             498
    unit = db.Column(db.String(32), nullable=True)                  499
    # 单位                                                          500
```

```
    ideal_value = db.Column(db.String(64), nullable=True)
    # 理想值（可配置）
    ideal_type = db.Column(db.String(32), nullable=True)
    # benefit/cost/mid/discrete
    # 启用该指标的微地质单元（逗号分隔）。岩质边坡不启用土壤指标、平台不启用地形指标
    enabled_units = db.Column(db.String(128), nullable=True)
    description = db.Column(db.Text, nullable=True)
    # 取值依据说明
    def units_list(self):
        return [s for s in (self.enabled_units or
    "").split(",") if s]
class WeightScheme(db.Model):
    """权重方案：按微地质单元类型分别保存。"""
    __tablename__ = "weight_schemes"
    id = db.Column(db.Integer, primary_key=True)
    unit_type = db.Column(db.String(32), nullable=False)
    # 所属单元类型
    name = db.Column(db.String(64), nullable=True)
    # 方案名称
    level = db.Column(db.Integer, nullable=False,
    default=1)          # 1=一级指标 2=二级指标
    is_default = db.Column(db.Boolean, default=False)
    # 是否默认（论文表 3）
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class WeightItem(db.Model):
    """权重明细：方案内的各项权重。"""
    __tablename__ = "weight_items"
    id = db.Column(db.Integer, primary_key=True)
    scheme_id = db.Column(db.Integer,
    db.ForeignKey("weight_schemes.id"), nullable=False)
    indicator_code = db.Column(db.String(32),
    nullable=False)        # 指标编码
    weight = db.Column(db.Float, nullable=False)
    # 权重值
class ExpertMatrix(db.Model):
    """专家判断矩阵：多专家打分的原始矩阵。"""
    __tablename__ = "expert_matrices"
    id = db.Column(db.Integer, primary_key=True)
    scheme_id = db.Column(db.Integer,
    db.ForeignKey("weight_schemes.id"), nullable=True)
    # 直接标注适用对象，便于无需先建权重方案即可录入专家矩阵
    unit_type = db.Column(db.String(32), nullable=True)
    # 微地质单元类型
    l1_name = db.Column(db.String(64), nullable=True)
    # 所属一级指标（二级矩阵用）
    expert_name = db.Column(db.String(64), nullable=True)
    # 专家姓名
    level = db.Column(db.Integer, nullable=False,
    default=1)          # 矩阵对应指标层级
```

```
    matrix_json = db.Column(db.Text, nullable=False)
    # 判断矩阵（JSON 二维数组）
    cr = db.Column(db.Float, nullable=True)
    # 该专家矩阵的一致性比例
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class IndicatorData(db.Model):
    """指标实测数据：按微地质单元组织。"""
    __tablename__ = "indicator_data"
    id = db.Column(db.Integer, primary_key=True)
    unit_id = db.Column(db.Integer, db.ForeignKey("micro_uni
    ts.id"), nullable=False, unique=True)
    soil_ratio = db.Column(db.Float, nullable=True)
    # 土石比
    thickness = db.Column(db.Float, nullable=True)
    # 土层厚度 cm
    ph = db.Column(db.Float, nullable=True)
    organic_matter = db.Column(db.Float, nullable=True)
    # 有机质 g/kg
    # 重金属含量，JSON 字符串：{"Hg": 0.05, ...}，未检测项缺失或为 null
    heavy_metals = db.Column(db.Text, nullable=True)
    coverage = db.Column(db.Float, nullable=True)
    # 植被覆盖度 %
    shannon = db.Column(db.Float, nullable=True)
    # Shannon-Wiener 多样性指数
    slope = db.Column(db.Float, nullable=True)
    # 坡度 °
    aspect = db.Column(db.String(16), nullable=True)
    # 坡向：阳坡/阴阳坡/阴坡
    def heavy_metals_dict(self):
        """把重金属 JSON 解析成字典。
        字段是手工可改的文本，被脏写或导入半个 JSON 时不能让整页 500：
        解析失败按「未检测」处理（空字典），并把原文留在库里供排查。
        """
        if not self.heavy_metals:
            return {}
        try:
            data = json.loads(self.heavy_metals)
        except (ValueError, TypeError):
            return {}
        return data if isinstance(data, dict) else {}
class EvaluationResult(db.Model):
    """评价结果。"""
    __tablename__ = "evaluation_results"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    final_score = db.Column(db.Float, nullable=True)
    # 最终评价因子 F
    grade = db.Column(db.String(32), nullable=True)
```

```
    # 适宜性等级
    measures = db.Column(db.Text, nullable=True)
    # 推荐措施
    unit_scores = db.Column(db.Text, nullable=True)
    # 各单元得分 JSON
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class ScoreDetail(db.Model):
    """得分明细：每一层指标的权重、标准化值、加权贡献，供追溯与复核。"""
    __tablename__ = "score_details"
    id = db.Column(db.Integer, primary_key=True)
    result_id = db.Column(db.Integer,
    db.ForeignKey("evaluation_results.id"), nullable=False)
    unit_id = db.Column(db.Integer, nullable=True)
    l1_name = db.Column(db.String(64), nullable=True)
    # 一级指标
    l2_name = db.Column(db.String(64), nullable=True)
    # 二级指标
    raw_value = db.Column(db.String(64), nullable=True)
    # 实测值
    std_value = db.Column(db.Float, nullable=True)
    # 标准化值
    weight = db.Column(db.Float, nullable=True)
    # 权重
    contribution = db.Column(db.Float, nullable=True)
    # 加权贡献
# ==== app/forms.py ====
# -*- coding: utf-8 -*-
"""表单定义（共享，由甲统一维护）。
各模块的 Flask-WTF 表单集中于此。
"""
import re
from flask_wtf import FlaskForm
from wtforms import SelectField, SubmitField, StringField,
    PasswordField, BooleanField
from wtforms.validators import (DataRequired, Length,
    Optional, EqualTo,
                                ValidationError)
from app.utils.constants import UNIT_TYPES, ASPECT_TYPES,
    GENDERS
# 邮箱格式：够用的轻量校验。
# WTForms 自带的 Email() 需要额外安装 email_validator,
# 为一个字段引入依赖不划算，这里自己用正则判断。
EMAIL_RE = re.compile(r"^[^@\s]+@[^@\s]+\.[A-Za-z]{2,}$")
def email_format(form, field):
    v = (field.data or "").strip()
    if v and not EMAIL_RE.match(v):
        raise ValidationError("邮箱格式不正确")
def email_ok(value):
    """邮箱格式判断，供不用 WTForms 的表单复用（如管理员编辑用户）。"""
```

```
    v = (value or "").strip()
    return (not v) or bool(EMAIL_RE.match(v))
class AHPSelectForm(FlaskForm):
    """选择微地质单元类型与指标层级，以确定判断矩阵规模。"""
    unit_type = SelectField("微地质单元类型",
    validators=[DataRequired()])
    level = SelectField("指标层级",
                        choices=[("1", "一级指标"), ("2",
    "二级指标")],
                        validators=[DataRequired()])
    l1 = SelectField("一级指标", validators=[],
    validate_choice=False)  # 动态填充，层级=2 时使用
    submit = SubmitField("生成判断矩阵")
    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)
        self.unit_type.choices = [(t, t) for t in
    UNIT_TYPES]
class EmptyForm(FlaskForm):
    """仅用于承载 CSRF 令牌的空表单（矩阵录入等自定义表单使用）。"""
    pass
class LoginForm(FlaskForm):
    """登录表单。支持「记住我」以在关闭浏览器后保持登录状态。
    校验消息用中文写明：登录框带 novalidate，浏览器不会代填提示，
    字段为空时若不自己给消息，用户点了「登录」会以为没反应。
    """
    username = StringField("账户名称",
                           validators=[DataRequired(message=
    "请输入账户名称"),
                                       Length(max=64,

    message="账户名称不能超过 64 个字符")])
    password = PasswordField("密码",
                             validators=[DataRequired(messag
    e="请输入密码")])
    remember = BooleanField("记住我")
    submit = SubmitField("登录")
GENDER_CHOICES = [("", "不填")] + [(g, g) for g in GENDERS]
class ProfileForm(FlaskForm):
    """个人资料：昵称、性别与联系方式。字段都可留空。
    职务不在这里 —— 只能由管理员在「个人设置 · 账号」里改。
    """
    display_name = StringField("昵称", validators=[Optional(),
    Length(max=64)])
    gender = SelectField("性别", choices=GENDER_CHOICES,
    validators=[Optional()])
    email = StringField("邮箱", validators=[Optional(),
    email_format(),
                                            Length(max=128)])
    phone = StringField("手机号", validators=[Optional(),
    Length(max=32)])
```

```
    submit_info = SubmitField("保存资料")
class RegisterForm(FlaskForm):
    """注册：账号与密码必填，其余选填。注册后默认是操作员。"""
    username = StringField("账号名称",
                           validators=[DataRequired("请填写账号名称"),
                                       Length(min=2, max=64,
                                              message="账号名称2–64 个字符")])
    password = PasswordField("密码",
                             validators=[DataRequired("请设置密码"),
                                         Length(min=6,
                                                max=64,
                                                message="密码至少 6 位")])
    confirm = PasswordField("确认密码",
                            validators=[DataRequired(),
                                        EqualTo("password",
                                                message="两次输入的密码不一致")])
    display_name = StringField("昵称", validators=[Optional(),
    Length(max=64)])
    gender = SelectField("性别", choices=GENDER_CHOICES,
    validators=[Optional()])
    email = StringField("邮箱", validators=[Optional(),
    email_format,
                                            Length(max=128)])
    phone = StringField("手机号", validators=[Optional(),
    Length(max=32)])
    submit_reg = SubmitField("注册并登录")
class PasswordForm(FlaskForm):
    """修改密码：需要先验证当前密码。"""
    old_password = PasswordField("当前密码",
    validators=[DataRequired()])
    new_password = PasswordField("新密码",
                                 validators=[DataRequired(),
    Length(min=6, max=64,
                                             message="新密码至少 6 位")])
    confirm = PasswordField("确认新密码",
                            validators=[DataRequired(),
    EqualTo("new_password",
                                            message="两次输入的新密码不一致")])
    submit_pwd = SubmitField("修改密码")
# ==== app/utils/constants.py ====
# -*- coding: utf-8 -*-
"""评价指标常量表。
```

```
全部数值取自论文，作为系统计算的唯一基准：
- 论文 ４．２．２ 节：指标标准化理想极值与取值依据
- 论文 ４．２．３ 节 / 表 ４：适宜性等级划分
- 论文表 ３：按微地质单元分别确定的一级、二级指标权重
- 论文 ４．２．１ 节：平均随机一致性指标 ＲＩ
"""
# ----------------------------------------------------------------------
    ----------
# 一、微地质单元类型
# ----------------------------------------------------------------------
    ----------
# 论文 ４．１ 节："按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡
# 三种地质微单元类型"
UNIT_TYPES = ("岩质边坡", "碎石边坡", "平台")
# 每个单元类型启用的指标类别（论文表 ３ 隐含：岩质边坡无土壤指标，平台无地形指标）
# 供数据录入界面与评价引擎判断"该单元是否需要某类指标"
UNIT_ENABLED_INDICATORS = {
    "岩质边坡": ("植被指标", "地形指标"),                    # 不设土壤指标
    "碎石边坡": ("土壤指标", "植被指标", "地形指标"),
    "平台": ("土壤指标", "植被指标"),                        # 不设地形指标
}
# ----------------------------------------------------------------------
    ----------
# 二、指标标准化理想极值（论文 ４．２．２ 节）
# ----------------------------------------------------------------------
    ----------
IDEAL_SOIL_RATIO = 0.8          # 土石比理想值
IDEAL_THICKNESS = 20.0          # 土层厚度理想值 （ｃｍ）
IDEAL_PH_RANGE = (5.0, 9.0)     # ｐＨ 理想区间（适中型指标）
IDEAL_ORGANIC = 30.0            # 有机质理想值
    （ｇ/ｋｇ），全国第二次土壤普查养分等级划分标准二等最低值
IDEAL_COVERAGE = 75.0           # 植被覆盖度参照生态系统 （％）
IDEAL_SHANNON = 0.7             # Ｓｈａｎｎｏｎ-Ｗｉｅｎｅｒ 多样性指数参照生态系统
# 重金属极值 （ｍｇ/ｋｇ），《绿化种植土壤》ＣＪ/Ｔ ３４０—２０１６ 土壤重金属含量 ＩＶ 级最高值
# 论文原文：镉、汞、铅、铬、砷、镍、铜、锌分别为 2、2、530、400、45、220、600、800
HEAVY_METALS = ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu",
    "Zn")
HEAVY_METAL_LIMIT = {
    "Cd": 2.0, "Hg": 2.0, "Pb": 530.0, "Cr": 400.0,
    "As": 45.0, "Ni": 220.0, "Cu": 600.0, "Zn": 800.0,
}
# 污染指标下 8 种重金属"重要性相当，权重平均分配"（论文 ４．２．１ 节）
HEAVY_METAL_EQUAL_WEIGHT = 1.0 / 8.0
# 坡度分界，论文 ４．２．２ 节引《新一轮退耕还林还草总体方案》发改西部〔２０１４〕１７７２ 号
SLOPE_BREAK = 25.0
# 坡向离散赋值，论文 ４．２．２ 节："阳坡、阴阳坡、阴坡分别赋值 0．2、0．6、1"
ASPECT_SCORES = {"阳坡": 0.2, "阴阳坡": 0.6, "阴坡": 1.0}
ASPECT_TYPES = ("阳坡", "阴阳坡", "阴坡")
# ｐＨ 适中型标准化的容差边界（区间外线性衰减，避免得分突变）
PH_TOL_LOW = 4.0
```

```
PH_TOL_HIGH = 10.0
# ---------------------------------------------------------------------
    ----------------
# 三、适宜性等级划分（论文表 4）
# ---------------------------------------------------------------------
    ----------------
GRADE_LEVELS = (
    {"grade": "不适宜", "min": None, "max": 0.3,
     "meaning": "存在限制性条件或区域生态本底较差，自然恢复影响因子严重制约自然恢复",
     "measure": "生态重建、人工辅助"},
    {"grade": "基本适宜", "min": 0.3, "max": 0.7,
     "meaning": "生态本底良好，核心自然恢复影响因子基本满足自然恢复需求",
     "measure": "自然恢复为主，可辅以封闭场地和少量补种、补植、飞播等"},
    {"grade": "适宜", "min": 0.7, "max": None,
     "meaning": "各自然恢复影响因子均能满足自然恢复需求，自然演替速度快",
     "measure": "自然恢复"},
)
def grade_of(score):
    """按论文表 4 判定适宜性等级。"""
    for g in GRADE_LEVELS:
        lo, hi = g["min"], g["max"]
        if lo is None and score < hi:
            return g
        if hi is None and score >= lo:
            return g
        if lo is not None and hi is not None and lo <=
    score < hi:
            return g
    # score 恰好等于 0.7 时落入"适宜"
    return GRADE_LEVELS[-1]
# ---------------------------------------------------------------------
    ----------------
# 四、平均随机一致性指标 RI（论文 4.2.1 节给出 n=2,3,4,5，其余为常用扩展值）
# ---------------------------------------------------------------------
    ----------------
RI_TABLE = {1: 0.0, 2: 0.0, 3: 0.58, 4: 0.90, 5: 1.12, 6:
    1.24,
            7: 1.32, 8: 1.41, 9: 1.45, 10: 1.49}
# 一致性比例阈值（论文：当 CR<0.1 时，判断矩阵满足一致性要求）
CR_THRESHOLD = 0.1
# ---------------------------------------------------------------------
    ----------------
# 五、默认权重方案（论文表 3）
# ---------------------------------------------------------------------
    ----------------
# 论文 4.2.2 节："划分岩质边坡、碎石土边坡、平台三类微地质单元开展独立评价，
# 分别计算评价指标权重"（论文摘要、图 2 流程与表 5 均作"碎石边坡"，
# 系统统一采用"碎石边坡"）。权重由 10 位专家打分后经 AHP 方法确定。
# 下表为论文表 3 所载的归一化权重，作为系统默认方案。
DEFAULT_WEIGHTS = {
```

```
        "岩质边坡"：  {
            "ｌ１"：  {"植被指标"：  ０．４５，  "地形指标"：  ０．５５},
            "ｌ２"：  {
                "植被指标"：  {"植被覆盖度"：  ０．６７，  "生物多样性"：  ０．３３},
                "地形指标"：  {"坡度"：  ０．６４，  "坡向"：  ０．３６},
            },
        },
        "碎石边坡"：  {
            "ｌ１"：  {"土壤指标"：  ０．４４，  "植被指标"：  ０．３１，  "地形指标"：  ０．２５},
            "ｌ２"：  {
                "土壤指标"：  {"土石比"：  ０．２５，  "土层厚度"：  ０．２５，  "ｐＨ"：  ０．１７，
                              "有机质"：  ０．１７，  "污染指标"：  ０．１７},
                "植被指标"：  {"植被覆盖度"：  ０．６４，  "生物多样性"：  ０．３６},
                "地形指标"：  {"坡度"：  ０．６７，  "坡向"：  ０．３３},
            },
        },
        "平台"：  {
            "ｌ１"：  {"土壤指标"：  ０．５４，  "植被指标"：  ０．４６},
            "ｌ２"：  {
                "土壤指标"：  {"土石比"：  ０．２５，  "土层厚度"：  ０．２５，  "ｐＨ"：  ０．１７，
                              "有机质"：  ０．１７，  "污染指标"：  ０．１７},
                "植被指标"：  {"植被覆盖度"：  ０．６１，  "生物多样性"：  ０．３９},
            },
        },
}
#  - - -
```

```
"""
from sqlalchemy import text
#  （表名， 列名， 列定义）
REQUIRED_COLUMNS = [
    #  遥感影像挂接
    ("mine_patches", "image_path", "VARCHAR(256)"),
    ("mine_patches", "image_note", "VARCHAR(256)"),
    ("mine_patches", "image_uploaded_at", "DATETIME"),
    #  指标维护
    ("indicators", "enabled_units", "VARCHAR(128)"),
    ("indicators", "description", "TEXT"),
    #  多专家矩阵
    ("expert_matrices", "unit_type", "VARCHAR(32)"),
    ("expert_matrices", "l1_name", "VARCHAR(64)"),
    ("expert_matrices", "cr", "FLOAT"),
    #  用户个人资料（账户页）
    ("users", "display_name", "VARCHAR(64)"),
    ("users", "email", "VARCHAR(128)"),
    ("users", "phone", "VARCHAR(32)"),
    ("users", "gender", "VARCHAR(8)"),
    ("users", "avatar", "VARCHAR(256)"),
]
#  职务体系由「管理员 / 评价员 / 查看者」三种收敛为「管理员 / 操作员」两种，
#  旧数据里的后两种统一归为操作员。
ROLE_MIGRATION = {"evaluator": "operator", "viewer":
    "operator"}
def _add_columns(db):
    added = []
    with db.engine.connect() as conn:
        for table, column, ddl in REQUIRED_COLUMNS:
            rows = conn.execute(text("PRAGMA
    table_info(%s)" % table)).fetchall()
            if not rows:
                continue                            #
    表还不存在，create_all 会建
            existing = {r[1] for r in rows}
            if column in existing:
                continue
            conn.execute(text(
                "ALTER TABLE %s ADD COLUMN %s %s" % (table,
    column, ddl)))
            conn.commit()
            added.append("%s.%s" % (table, column))
    return added
def _migrate_roles(db):
    """把旧角色名统一为新的两种，返回改动的行数。"""
    changed = 0
    with db.engine.connect() as conn:
        rows = conn.execute(text("PRAGMA
    table_info(users)")).fetchall()
```

```
        if not rows:                                                    951
            return 0                                                    952
        for old, new in ROLE_MIGRATION.items():                         953
            r = conn.execute(text("UPDATE users SET role =              954
    :new "                                                              955
                                  "WHERE role = :old"),                 956
                                 {"new": new, "old": old})              957
            changed += r.rowcount or 0                                  958
        conn.commit()                                                   959
    return changed                                                      960
def ensure_schema(db):                                                  961
    """为已有表补齐缺失列，并做必要的数据迁移。返回变更清单。"""                    962
    try:                                                                963
        changed = _add_columns(db)                                      964
        n = _migrate_roles(db)                                          965
        if n:                                                           966
            changed.append("用户角色归一为管理员/操作员 %d 条" % n)              967
        return changed                                                  968
    except Exception as e:                                  #           969
    迁移失败不应阻断启动                                                  970
        print("[schema] 结构检查跳过：%s" % e)                             971
        return []                                                       972
# ==== app/utils/uploads.py ====                                        973
# -*- coding: utf-8 -*-                                                 974
"""上传文件处理（遥感影像、批量导入）。"""                                    975
import os                                                               976
import re                                                               977
import uuid                                                             978
from datetime import datetime                                           979
from werkzeug.utils import secure_filename                              980
def _safe_ext(filename):                                                981
    return os.path.splitext(filename)[1].lower().lstrip(".")            982
def ext_allowed(filename, allowed):                                     983
    return _safe_ext(filename) in allowed                               984
def save_upload(file_storage, subdir, allowed_ext,                      985
    upload_root):                                                       986
    """保存上传文件，返回相对 static 的路径（如 uploads/patches/xxx.png）。    987
    文件名重命名为「时间戳_随机串.扩展名」，避免中文名与重名问题。                 988
    """                                                                 989
    if file_storage is None or not file_storage.filename:               990
        return None, "未选择文件"                                        991
    ext = _safe_ext(file_storage.filename)                              992
    if ext not in allowed_ext:                                          993
        return None, "不支持的文件类型：.%s" % ext                        994
    target_dir = os.path.join(upload_root, subdir)                      995
    os.makedirs(target_dir, exist_ok=True)                              996
    # 仅保留安全字符，并加时间戳与随机串防重名                                997
    base = secure_filename(os.path.splitext(file_storage.fil            998
    ename)[0]) or "file"                                                999
    base = re.sub(r"[^A-Za-z0-9_\-]", "", base)[:40] or                1000
```

```
    "file"                                                                1001
    name = "%s_%s.%s" % (datetime.now().strftime("%Y%m%d%H%M1002
    %S"),                                                                 1003
                                uuid.uuid4().hex[:6], ext)                1004
    abs_path = os.path.join(target_dir, name)                             1005
    file_storage.save(abs_path)                                           1006
    rel = os.path.join("uploads", subdir,                                 1007
    name).replace("\\", "/")                                              1008
    return rel, None                                                      1009
def remove_upload(rel_path, upload_root):                                 1010
    """删除已保存的上传文件（相对 static 的路径）。"""                    1011
    if not rel_path:                                                      1012
        return                                                            1013
    abs_path = os.path.join(upload_root,                                  1014
    rel_path.replace("uploads/", "", 1))                                  1015
    try:                                                                  1016
        if os.path.isfile(abs_path):                                      1017
            os.remove(abs_path)                                           1018
    except OSError:                                                       1019
        pass                                                              1020
# ==== app/utils/charts.py ====                                           1021
# -*- coding: utf-8 -*-                                                   1022
"""轻量图表。                                                            1023
用纯 SVG 输出，不依赖任何绘图库或前端脚本：                              1024
- 服务端渲染，页面与打印（PDF）中都能稳定显示                            1025
- 每个图元都带数值标签，黑白打印时仍可读（不单靠颜色区分）              1026
- 单色方案，符合界面「一个视图一个强调色」的约定                        1027
"""                                                                       1028
from html import escape                                                   1029
ACCENT = "#1f3a5f"                                                        1030
ACCENT_SOFT = "#7f93ad"                                                   1031
INK = "#23262b"                                                           1032
MUTED = "#6b7280"                                                         1033
LINE = "#dfe3e8"                                                          1034
# 灰度梯度：黑白打印时相邻扇区 / 条柱仍可区分                            1035
GRAYS = ["#1f3a5f", "#456188", "#6d86a8", "#96a9c4",                      1036
    "#bcc9da", "#dbe3ec"]                                                 1037
def _fmt(v, digits=4):                                                    1038
    try:                                                                  1039
        return ("%%.%df" % digits) % float(v)                             1040
    except (TypeError, ValueError):                                       1041
        return str(v)                                                     1042
def hbars(items, width=620, bar_h=22, gap=12, label_w=150,                1043
    value_w=76):                                                          1044
    """水平条形图。items: [(标签, 数值, 显示文本)]"""                     1045
    if not items:                                                         1046
        return '<p class="muted">暂无数据</p>'                            1047
    n = len(items)                                                        1048
    height = n * (bar_h + gap) + 10                                       1049
    max_v = max((abs(float(v)) for _, v, _ in items),                     1050
```

```
    default=1) or 1                                                    1051
    body_w = width - label_w - value_w                                 1052
    parts = ['<svg viewBox="0 0 %d %d" width="100%%"                   1053
    height="%d" '                                                      1054
              'role="img" aria-label="条形图" '                        1055
              'style="max-width:%dpx;font-family:inherit;">'           1056
    % (width, height, height, width)]                                  1057
    for i, (label, value, disp) in enumerate(items):                   1058
        y = i * (bar_h + gap) + 5                                      1059
        frac = (abs(float(value)) / max_v) if max_v else 0             1060
        w = max(2.0, body_w * frac)                                    1061
        fill = GRAYS[i % len(GRAYS)]                                   1062
        parts.append(                                                  1063
            '<text x="%d" y="%d" text-anchor="end"                     1064
    font-size="12" fill="%s">%s</text>'                                1065
            % (label_w - 10, y + bar_h * 0.72, INK,                    1066
    escape(str(label))))                                               1067
        parts.append(                                                  1068
            '<rect x="%d" y="%d" width="%.1f" height="%d"              1069
    rx="3" fill="%s"/>'                                                1070
            % (label_w, y, w, bar_h, fill))                            1071
        parts.append(                                                  1072
            '<text x="%d" y="%d" font-size="12" fill="%s" '            1073
            'style="font-variant-numeric:tabular-nums;">%s</           1074
    text>'                                                             1075
            % (label_w + w + 8, y + bar_h * 0.72, MUTED,               1076
    escape(str(disp))))                                                1077
    parts.append('</svg>')                                             1078
    return "".join(parts)                                              1079
def donut(items, size=210, thickness=34):                              1080
    """环形图（权重分布）。items: [(标签, 数值, 显示文本)]                1081
    图例中给出数值与占比，黑白打印时靠图例与灰度深浅区分。                  1082
    """                                                                1083
    vals = [(l, float(v), d) for l, v, d in items if                   1084
    float(v) > 0]                                                      1085
    if not vals:                                                       1086
        return '<p class="muted">暂无数据</p>'                          1087
    total = sum(v for _, v, _ in vals) or 1                            1088
    r = (size - thickness) / 2                                         1089
    cx = cy = size / 2                                                 1090
    import math                                                        1091
    parts = ['<svg viewBox="0 0 %d %d" width="%d"                      1092
    height="%d" role="img" '                                           1093
             'aria-label="权重分布环形图">' % (size, size, size,       1094
    size)]                                                             1095
    angle = -math.pi / 2                                               1096
    for i, (label, v, disp) in enumerate(vals):                        1097
        sweep = 2 * math.pi * (v / total)                              1098
        a0, a1 = angle, angle + sweep                                  1099
        x0, y0 = cx + r * math.cos(a0), cy + r *                       1100
```

```
    math.sin(a0)                                                    1101
        x1, y1 = cx + r * math.cos(a1), cy + r *                    1102
    math.sin(a1)                                                    1103
        large = 1 if sweep > math.pi else 0                         1104
        parts.append(                                               1105
            '<path d="M %.2f %.2f A %.2f %.2f 0 %d 1 %.2f           1106
    %.2f" '                                                         1107
            'fill="none" stroke="%s" stroke-width="%d"/>'           1108
           % (x0, y0, r, r, large, x1, y1, GRAYS[i %                1109
    len(GRAYS)], thickness))                                        1110
        angle = a1                                                  1111
    parts.append('<text x="%d" y="%d" text-anchor="middle"          1112
    font-size="12" '                                                1113
                 'fill="%s">合计</text>' % (cx, cy - 2,             1114
    MUTED))                                                         1115
    parts.append('<text x="%d" y="%d" text-anchor="middle"          1116
    font-size="15" '                                                1117
                 'fill="%s" style="font-variant-numeric:tabu        1118
    lar-nums;">%s</text>'                                           1119
                 % (cx, cy + 18, INK, _fmt(total, 4)))              1120
    parts.append('</svg>')                                          1121
    legend = ['<table style="width:auto;margin-top:10px;"><t        1122
    head><tr>'                                                      1123
              '<th scope="col">项目</th><th                         1124
    scope="col">数值</th>'                                          1125
              '<th scope="col">占比</th></tr></thead><tbody>']      1126
    for i, (label, v, disp) in enumerate(vals):                     1127
        legend.append(                                              1128
            '<tr><td style="text-align:left;">'                     1129
            '<span style="display:inline-block;width:10px;he        1130
    ight:10px;'                                                     1131
            'border-radius:2px;background:%s;margin-right:6p        1132
    x;"></span>%s</td>'                                             1133
            '<td class="mono">%s</td><td                            1134
    class="mono">%.1f%%</td></tr>'                                  1135
           % (GRAYS[i % len(GRAYS)], escape(str(label)),            1136
    escape(str(disp)),                                              1137
               100.0 * v / total))                                  1138
    legend.append('</tbody></table>')                               1139
    return ('<div style="display:flex;flex-wrap:wrap;gap:20p        1140
    x;align-items:center;">'                                        1141
            + "".join(parts) + "".join(legend) + '</div>')          1142
# ==== app/utils/browser.py ====                                    1143
# -*- coding: utf-8 -*-                                             1144
"""HTML 转 PDF（报告导出）。                                        1145
复用本机已安装的 Chrome/Edge 无头模式打印，避免引入 reportlab、     1146
weasyprint 等额外依赖。生成过程使用系统临时目录中的 ASCII 文件名，  1147
避免中文路径在 file:// URL 下出问题。                               1148
"""                                                                 1149
import os                                                           1150
```

```
import shutil
import subprocess
import tempfile
import uuid
CANDIDATES = [
    r"C:\Program Files\Google\Chrome\Application\chrome.exe",
    r"C:\Program Files (x86)\Google\Chrome\Application\chrome.exe",
    r"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe",
    r"C:\Program Files\Microsoft\Edge\Application\msedge.exe",
]
# 允许通过环境变量覆盖，便于在其他机器上指定浏览器
ENV_KEY = "DSH_BROWSER_PATH"
def browser_path():
    """定位可用的浏览器可执行文件，找不到返回 None。"""
    override = os.environ.get(ENV_KEY)
    if override and os.path.isfile(override):
        return override
    for p in CANDIDATES:
        if os.path.isfile(p):
            return p
    for name in ("chrome", "msedge", "chromium"):
        found = shutil.which(name)
        if found:
            return found
    return None
def html_to_pdf(html, timeout=90):
    """把 HTML 字符串渲染为 PDF，返回字节；失败返回 (None, 原因)。"""
    exe = browser_path()
    if exe is None:
        return None, ("未找到可用的浏览器（Chrome/Edge），无法生成 PDF。"
                      "可设置环境变量 %s 指向浏览器可执行文件。" % ENV_KEY)
    tmpdir = tempfile.mkdtemp(prefix="dshpdf_")
    tag = uuid.uuid4().hex[:8]
    html_path = os.path.join(tmpdir, "r_%s.html" % tag)
    pdf_path = os.path.join(tmpdir, "r_%s.pdf" % tag)
    try:
        with open(html_path, "w", encoding="utf-8") as f:
            f.write(html)
        cmd = [
            exe, "--headless=new", "--disable-gpu",
    "--no-sandbox",
            "--no-first-run", "--no-default-browser-check",
            "--no-pdf-header-footer",
            "--virtual-time-budget=4000",
            "--print-to-pdf=%s" % pdf_path,
            "file:///" + html_path.replace("\\", "/"),
```

```
            ]
            p = subprocess.run(cmd, capture_output=True,
      text=True,
                               timeout=timeout,
      encoding="utf-8", errors="replace")
            if not os.path.isfile(pdf_path) or
      os.path.getsize(pdf_path) < 800:
                return None, "浏览器未能生成 PDF（%s）" % (p.stderr or
      "").strip()[:200]
            with open(pdf_path, "rb") as f:
                return f.read(), None
      except subprocess.TimeoutExpired:
            return None, "生成 PDF 超时，请稍后重试"
      except Exception as e:
            return None, "生成 PDF 失败：%s" % e
      finally:
            shutil.rmtree(tmpdir, ignore_errors=True)
# ==== app/services/ahp_service.py ====
# -*- coding: utf-8 -*-
"""AHP 权重引擎。
严格按论文 4.2.1 节实现：
- 式(2)(3)(4)：特征根法求权重（行元素乘积 → n 次方根 → 归一化）
- 式(5)(6)(7)：一致性检验（λmax、CI、CR）
- 论文要求 CR < 0.1；RI 按阶数查表
"""
import numpy as np
from app.utils.constants import RI_TABLE, CR_THRESHOLD
def matrix_from_upper(upper):
      """由判断矩阵上三角构造完整互反矩阵。
      参数
      ----
      upper : list[list[float]] 或 np.ndarray，上三角元素（不含对角线）。
            例：3 阶时传入 [[a12, a13], [a23]]。
            对角线为 1，下三角取倒数（a_ji = 1/a_ij）。
      """
      upper = np.asarray(upper, dtype=float)
      k = upper.shape[0]
      n = int(round((1 + (1 + 8 * k) ** 0.5) / 2))
      A = np.ones((n, n), dtype=float)
      idx = 0
      for i in range(n):
            for j in range(i + 1, n):
                A[i, j] = upper[idx]
                A[j, i] = 1.0 / upper[idx]
                idx += 1
      return A
def weights_and_consistency(A):
      """由判断矩阵 A 计算权重与一致性指标。
      返回
      ----
```

```
    dict：{weights: np.ndarray, lambda_max, ci, cr, n,
    consistent: bool}
    """
    A = np.asarray(A, dtype=float)
    n = A.shape[0]
    # 式(2) 行元素乘积
    row_prod = np.prod(A, axis=1)
    # 式(3) 开 n 次方根
    root = row_prod ** (1.0 / n)
    # 式(4) 归一化
    weights = root / root.sum()
    # 式(5) 最大特征值 λmax = (1/n) Σ (A·W)_i / W_i
    aw = A.dot(weights)
    lambda_max = float(np.sum(aw / weights) / n)
    # 式(6) 一致性指标 CI
    ci = (lambda_max - n) / (n - 1) if n > 1 else 0.0
    # 式(7) 一致性比例 CR = CI / RI
    ri = RI_TABLE.get(n, 1.49)
    cr = ci / ri if ri else 0.0
    return {
        "weights": weights,
        "lambda_max": lambda_max,
        "ci": ci,
        "cr": cr,
        "n": n,
        "consistent": cr < CR_THRESHOLD,
    }
def find_most_inconsistent(A):
    """定位判断矩阵中最不一致的一对比较。
    依据：完全一致矩阵应满足 a_ij = w_i / w_j。
    计算每个上三角元素与其理想值的相对偏差，返回偏差最大者。
    返回
    ----
    dict：{i, j, value, expected, deviation}，偏差最大的 (i, j)
    及对应取值。
    """
    A = np.asarray(A, dtype=float)
    r = weights_and_consistency(A)
    w = r["weights"]
    n = A.shape[0]
    best = None
    for i in range(n):
        for j in range(i + 1, n):
            expected = w[i] / w[j]
            if expected == 0:
                continue
            deviation = abs(A[i, j] - expected) / expected
            if best is None or deviation >
    best["deviation"]:
                best = {"i": i, "j": j, "value": A[i, j],
```

```
                                "expected": expected, "deviation":  1301
    deviation}                                                      1302
    return best                                                     1303
def aggregate_matrices(matrices):                                   1304
    """多专家判断矩阵聚合：几何平均（保持互反性）。                  1305
    参数                                                            1306
    ----                                                            1307
    matrices : list[np.ndarray]，多个同阶判断矩阵。                 1308
    """                                                             1309
    stack = np.stack([np.asarray(m, dtype=float) for m in           1310
    matrices])                                                      1311
    return np.prod(stack, axis=0) ** (1.0 / len(stack))             1312
def judge_from_weights(weights):                                    1313
    """由权重向量构造完全一致判断矩阵并检验。                        1314
    用途：验证给定权重向量能否作为某个完全一致判断矩阵的主特征向量  1315
    （A_ij = w_i / w_j 时 λmax = n、CR = 0）。                       1316
    """                                                             1317
    w = np.asarray(weights, dtype=float)                            1318
    n = w.shape[0]                                                  1319
    A = np.outer(w, 1.0 / w)                                        1320
    r = weights_and_consistency(A)                                  1321
    r["matrix"] = A                                                 1322
    return r                                                        1323
# ==== app/services/standardization.py ====                       1324
# -*- coding: utf-8 -*-                                           1325
"""指标标准化（的一部分）。                                        1326
按论文 4.2.2 节式(8) 实现：                                        1327
    x' = (x - x_min) / (x_max - x_min)                              1328
按指标性质分为四类：越大越优 / 越小越优 / 适中 / 离散赋值。        1329
所有理想极值取自 app.utils.constants，可配置。                     1330
"""                                                                 1331
from app.utils.constants import (                                   1332
    IDEAL_SOIL_RATIO, IDEAL_THICKNESS, IDEAL_PH_RANGE,              1333
    IDEAL_ORGANIC,                                                  1334
    IDEAL_COVERAGE, IDEAL_SHANNON, HEAVY_METAL_LIMIT,               1335
    HEAVY_METAL_EQUAL_WEIGHT, SLOPE_BREAK, ASPECT_SCORES,           1336
    PH_TOL_LOW, PH_TOL_HIGH,                                        1337
)                                                                   1338
def std_benefit(value, ideal):                                      1339
    """越大越优指标标准化：以理想值为上限，按比例给分，饱和于 [0,1]。"""  1340
    if value is None:                                               1341
        return None                                                 1342
    return max(0.0, min(1.0, value / ideal))                        1343
def std_cost(value, extreme):                                       1344
    """越小越优指标标准化：以极值为参考，线性折减至 [0,1]。"""      1345
    if value is None:                                               1346
        return None                                                 1347
    return max(0.0, min(1.0, 1.0 - value / extreme))                1348
def std_mid_ph(value, lo, hi, tol_lo=PH_TOL_LOW,                    1349
    tol_hi=PH_TOL_HIGH):                                            1350
```

```
    """适中型指标标准化（pH）：理想区间内取 1，区间外线性衰减。"""
    if value is None:
        return None
    if lo <= value <= hi:
        return 1.0
    if value > hi:
        return max(0.0, (tol_hi - value) / (tol_hi - hi))
    return max(0.0, (value - tol_lo) / (lo - tol_lo))
def std_slope(value, break_point=SLOPE_BREAK, upper=90.0):
    """坡度标准化。
    论文 4.2.2 节：以 25° 为陡坡重要分界。25° 及以下记满分 1，
    超过 25° 线性折减，90° 记 0。
    """
    if value is None:
        return None
    if value <= break_point:
        return 1.0
    return max(0.0, (upper - value) / (upper - break_point))
def aspect_score(aspect):
    """坡向离散赋值：阳坡 0.2 / 阴阳坡 0.6 / 阴坡 1.0（论文 4.2.2 节）。"""
    if aspect is None:
        return None
    return ASPECT_SCORES.get(aspect)
def pollution_score(heavy_metals, limits=None):
    """污染指标：8 种重金属等权（各 0.125）合成。
    论文注：未检测 Cr、Zn 的，从矿山类型看未受污染，按无污染（=1）处理。
    heavy_metals : dict，键为金属名，值为含量 mg/kg；缺失键视为未检测。
    limits : 可选，金属 → 极值；不传则用代码常量（界面改过就传库里的值）。
    """
    table = limits or HEAVY_METAL_LIMIT
    total = 0.0
    for metal, extreme in table.items():
        value = heavy_metals.get(metal)
        total += std_cost(value, extreme) if value is not
    None else 1.0
    return total / len(table)
def standardize_unit(unit_type, data, weights=None,
    settings=None):
    """对单个微地质单元的全部指标做标准化。
    参数
    ----
    unit_type : str，岩质边坡 / 碎石边坡 / 平台
    data : dict，实测数据，字段见接口契约：
        soil_ratio, thickness, ph, organic_matter,
    heavy_metals(dict),
        coverage, shannon, slope, aspect
    weights : dict，该单元类型的权重（app.utils.constants.DEFAULT_WEIG
    HTS 结构），
                未提供时用默认值。
    settings : app.services.unit_config.UnitSettings，图斑级的基准值
```

```
与理想取值。
        不传则全部按代码常量 —— 历史数据与验证算例走这条路径，结果不变。
返回
----
dict：{ 一级指标： { 二级指标： （实测值， 标准化值， 权重， 贡献） } }
标准化值可能为 None，表示该指标不适用或方法待定。
"""
from app.utils.constants import DEFAULT_WEIGHTS
from app.services import unit_config as uc
if weights is None:
    weights = DEFAULT_WEIGHTS[unit_type]
if settings is None:
    settings = uc.UnitSettings()
# 理想取值：库里有数字就用库里的，否则用常量
def ideal(code, fallback):
    return settings.ideal(code, fallback)
l1_enabled = weights["l1"].keys()
l2w = weights["l2"]
result = {}
# 一级指标是否启用：库里配过就按库里的（界面上的「启用单元」），没配按常量
def group_on(l1_name, l1_code):
    return settings.group_enabled(unit_type, l1_name,
l1_code) is not False
# ---- 土壤指标 ----
if "土壤指标" in l1_enabled and group_on("土壤指标", "soil"):
    result["土壤指标"] = {
        "土石比": (data.get("soil_ratio"),
                   std_benefit(data.get("soil_ratio"),
                               ideal("soil_ratio",
IDEAL_SOIL_RATIO)),
                   l2w["土壤指标"]["土石比"]),
        "土层厚度": (data.get("thickness"),
                     std_benefit(data.get("thickness"),
                                 ideal("thickness",
IDEAL_THICKNESS)),
                     l2w["土壤指标"]["土层厚度"]),
        "pH": (data.get("ph"),
               std_mid_ph(data.get("ph"),
*IDEAL_PH_RANGE),
               l2w["土壤指标"]["pH"]),
        "有机质": (data.get("organic_matter"),
                   std_benefit(data.get("organic_matter")
,
                               ideal("organic_matter",
IDEAL_ORGANIC)),
                   l2w["土壤指标"]["有机质"]),
        "污染指标": (data.get("heavy_metals") or {},
                     pollution_score(data.get("heavy_meta
ls") or {},
```

```
        uc.metal_limits(settings)),                                          1451
                                    l2w["土壤指标"]["污染指标"]),              1452
            }                                                                1453
        #  ----  植被指标  ----                                               1454
        if  "植被指标"  in  l1_enabled  and  group_on("植被指标",  "veg"):     1455
            result["植被指标"]  =  {                                         1456
                "植被覆盖度":  (data.get("coverage"),                        1457
                                    std_benefit(data.get("coverage"),        1458
                                                ideal("coverage",            1459
        IDEAL_COVERAGE)),                                                    1460
                                    l2w["植被指标"]["植被覆盖度"]),           1461
                "生物多样性":  (data.get("shannon"),                         1462
                                    std_benefit(data.get("shannon"),         1463
                                                ideal("shannon",             1464
        IDEAL_SHANNON)),                                                     1465
                                    l2w["植被指标"]["生物多样性"]),           1466
            }                                                                1467
        #  ----  地形指标  ----                                               1468
        if  "地形指标"  in  l1_enabled  and  group_on("地形指标",  "terrain"):1469
            result["地形指标"]  =  {                                         1470
                "坡度":  (data.get("slope"),                                 1471
                          std_slope(data.get("slope")),                      1472
                          l2w["地形指标"]["坡度"]),                          1473
                "坡向":  (data.get("aspect"),                                1474
                          aspect_score(data.get("aspect")),                  1475
                          l2w["地形指标"]["坡向"]),                          1476
            }                                                                1477
        return  result                                                       1478
#  ====  app/services/weight_service.py  ====                                1479
#  -*-  coding:  utf-8  -*-                                                  1480
"""权重方案的存取与生效权重解析。                                            1481
系统有两套权重来源：                                                        1482
1．  论文表  3  给出的默认权重（app.utils.constants.DEFAULT_WEIGHTS），     1483
    是**权威取值来源**，案例复算与报告结论以此为准；                        1484
2．  通过层次分析法在本系统中实际算出的权重，保存在                         1485
    weight_schemes  /  weight_items  两张表中，可标记为「生效」。           1486
本模块负责把这两者合成为计算引擎直接可用的嵌套权重结构：                    1487
以论文表  3  为底，用生效方案中已保存的条目覆盖。                          1488
没有任何生效方案时，即完全使用论文表  3  的权重。                          1489
这样既保证默认行为与论文一致，又使  AHP  计算结果真正参与评价，            1490
而不是算完即弃。                                                            1491
"""                                                                         1492
import  copy                                                                1493
from  app.models  import  db,  WeightScheme,  WeightItem                    1494
from  app.utils.constants  import  DEFAULT_WEIGHTS                          1495
from  app.services.indicator_service  import  TREE                          1496
#  方案名称：每个微地质单元类型固定一个可生效的方案                          1497
ACTIVE_NAME  =  "自定义权重方案（AHP  计算结果）"                           1498
def  _code_maps():                                                          1499
    """由指标树导出  编码  →  名称  的映射，避免名称在两处各写一遍。"""      1500
```

```
print("   全通过后显示「通过，可进入定量评价」：", "可进入定量评价" in body)
assert "可进入定量评价" in body
# 一项不通过
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "0",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("   一项不通过后显示「不通过」：", "不通过" in body)
print("   含失败项「区位约束」：", "区位约束" in body)
assert "不通过" in body
# 恢复为全部通过，避免把演示数据留在「不通过」状态影响后续步骤
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "1",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
print("   已恢复为全部通过：", "可进入定量评价" in r.get_data(as_text=True))
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
# ==== 测试/test_web_evaluate.py ====
# -*- coding: utf-8 -*-
"""端到端自检：录数据 → 执行评价 → 查看结果。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
# 与真实配置共用同一个 app.db，但关闭 CSRF 便于测试客户端
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
```

```
print(LINE)
print("测试 1  指标数据列表")
print(LINE)
r = client.get("/indicator/")
body = r.get_data(as_text=True)
print("  状态码：", r.status_code)
print("  含案例图斑：", "玉田县孤树镇白云岩采石场" in body)
print("  含三个单元：", all(n in body for n in ("岩质边坡", "碎石边坡",
    "平台")))
assert r.status_code == 200
assert "岩质边坡" in body and "碎石边坡" in body and "平台" in body
print()
print(LINE)
print("测试 2  指标数据录入页（按单元类型显示字段）")
print(LINE)
from app.models import MicroUnit, MinePatch
with app.app_context():
    unit_rock = MicroUnit.query.filter_by(unit_type="岩质边坡").
    first()
    unit_gravel = MicroUnit.query.filter_by(unit_type="碎石边坡"
    ).first()
r = client.get("/indicator/%d" % unit_rock.id)
b_rock = r.get_data(as_text=True)
print("  岩质边坡页：无「土石比」字段：", "土石比" not in b_rock, "（应 True）")
print("  岩质边坡页：含「植被覆盖度」：", "植被覆盖度" in b_rock)
assert "土石比" not in b_rock and "植被覆盖度" in b_rock
r = client.get("/indicator/%d" % unit_gravel.id)
b_gravel = r.get_data(as_text=True)
print("  碎石边坡页：含「土石比」字段：", "土石比" in b_gravel)
print("  碎石边坡页：含 8 种重金属：", all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn")))
assert "土石比" in b_gravel and all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn"))
print()
print(LINE)
print("测试 3  执行评价")
print(LINE)
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
r = client.post("/evaluate/%d" % patch.id, data={},
    follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转至结果页：", "适宜性评价结果" in body)
print("  含最终评价因子 F：", "最终评价因子 F" in body)
print("  含等级「适宜」：", "适宜" in body)
print("  含推荐措施「自然恢复」：", "自然恢复" in body)
assert "最终评价因子 F" in body and "自然恢复" in body
print()
print(LINE)
```

```
print("测试 4  结果持久化")
print(LINE)
from app.models import EvaluationResult
with app.app_context():
    res = EvaluationResult.query.order_by(EvaluationResult.id.desc()).first()
    print("  最终得分 F =", res.final_score)
    print("  等级 =", res.grade)
    assert res.grade == "适宜"
    assert 0.7 <= res.final_score < 1.0
print()
print(LINE)
print("端到端全部测试通过 √")
print(LINE)
# ==== 测试/test_web_report.py ====
# -*- coding: utf-8 -*-
"""报告输出自检：得分明细页 + Word 导出。"""
import os
import sys
import io
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
from app.models import MinePatch, EvaluationResult
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
    pid = patch.id
# 报告取自已保存的评价记录，因此先按网页路径真实执行一次评价
from app.views.evaluate import load_patch_input,
    save_evaluation
with app.app_context():
```

```
    units = load_patch_input(pid)
    rec = save_evaluation(pid, units)
    rid = rec.id
    print("前置：已执行评价并保存记录 id=%d，F=%.4f（%s）"
          % (rid, rec.final_score, rec.grade))
    from app.models import ScoreDetail
    n_detail = ScoreDetail.query.filter_by(result_id=rid).co
    unt()
    print("         逐层得分明细已落库：%d 条" % n_detail)
print()
print(LINE)
print("测试 1  报告列表")
print(LINE)
r = client.get("/report/")
print("  状态码：", r.status_code)
body = r.get_data(as_text=True)
print("  含「所属项目」列：", "所属项目" in body)
assert r.status_code == 200
print()
print(LINE)
print("测试 2  得分明细页")
print(LINE)
r = client.get("/report/%d" % pid)
body = r.get_data(as_text=True)
print("  状态码：", r.status_code)
print("  含「得分明细」：", "得分明细" in body)
print("  含「最终评价因子 F」：", "最终评价因子 F" in body)
print("  含二级指标明细（植被覆盖度 / 坡度 / 坡向）：", all(n in body for n in
    ("植被覆盖度", "坡度", "坡向")))
print("  含权重 / 标准化值列：", "标准化值" in body and "贡献" in body)
assert r.status_code == 200
assert "植被覆盖度" in body and "标准化值" in body
print()
print(LINE)
print("测试 3  Word 报告导出")
print(LINE)
r = client.get("/report/%d/word" % pid)
print("  状态码：", r.status_code)
print("  Content-Type:", r.headers.get("Content-Type",
    "")[:60])
data = r.data
print("  文件大小：", len(data), "字节")
print("  是否为有效 docx（PK 头）：", data[:2] == b"PK")
assert r.status_code == 200 and data[:2] == b"PK"
# 用 python-docx 打开验证内容
import docx
d = docx.Document(io.BytesIO(data))
text = "\n".join(p.text for p in d.paragraphs)
print("  含报告标题：", "自然恢复适宜性评价报告" in text)
print("  含评价结论章节：", "评价结论" in text)
```

```
print("   含得分明细章节：", "得分明细" in text)                    11579
print("   段落数：", len(d.paragraphs), " 表格数：", len(d.tables))  11580
assert "自然恢复适宜性评价报告" in text and len(d.tables) >= 3          11581
print()                                                            11582
print(LINE)                                                        11583
print("报告输出全部测试通过 √")                                      11584
print(LINE)                                                        11585
# ==== 测试/test_ui_fixes.py ====                                  11586
# -*- coding: utf-8 -*-                                            11587
"""界面修复自检：「记住我」与各模块删除功能。"""                          11588
import os                                                          11589
import re                                                          11590
import sys                                                         11591
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，                  11592
# 使 `from app import ...` 与 app/app.db 的定位不受影响。             11593
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil       11594
    e__)))                                                         11595
if ROOT not in sys.path:                                           11596
    sys.path.insert(0, ROOT)                                       11597
from app import create_app                                         11598
sys.stdout.reconfigure(encoding="utf-8")                           11599
LINE = "=" * 70                                                    11600
class TestConfig:                                                  11601
    SECRET_KEY = "test"                                            11602
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(         11603
        ROOT, "app", "app.db")                                     11604
    SQLALCHEMY_TRACK_MODIFICATIONS = False                         11605
    WTF_CSRF_ENABLED = False                                       11606
    TESTING = True                                                 11607
    REMEMBER_COOKIE_DURATION = __import__("datetime").timede       11608
    lta(days=30)                                                   11609
app = create_app(TestConfig)                                       11610
client = app.test_client()                                         11611
# ------------------------------------------------------------     11612
    ----------------                                               11613
print(LINE)                                                        11614
print("一、「记住我」")                                               11615
print(LINE)                                                        11616
html = client.get("/login").get_data(as_text=True)                 11617
# 1) 复选框必须有可点击关联的 label（这是原先失效的根因）                 11618
#    只判断「复选框与文字是否落在同一个 <label> 内」，不绑定具体类名 ——    11619
#    否则换个样式类就会误报，而真正的回归点其实与类名无关。               11620
labels = re.findall(r"<label[^>]*>(.*?)</label>", html,            11621
    re.S)                                                          11622
wrapped = any(('name="remember"' in b and "记住我" in b) for b     11623
    in labels)                                                     11624
print("   %s 「记住我」文字与复选框包在同一 label 内" % ("[符合]" if wrapped  11625
    else "[不符合]"))                                              11626
assert wrapped, "复选框未与文字关联，点击文字不会勾选"                    11627
has_for = 'for="remember"' in html                                 11628
```

```
print("  %s label 带 for 属性指向复选框" % ("[符合]" if has_for else
    "[不符合]"))
# 2) 勾选后应下发带过期时间的 remember_token
c = app.test_client()
r = c.post("/login", data={"username": "admin", "password":
    "admin123",
                           "remember": "y"},
    follow_redirects=False)
cookies = r.headers.getlist("Set-Cookie")
rt = [h for h in cookies if h.startswith("remember_token=")]
print("  %s 勾选后下发 remember_token" % ("[符合]" if rt else
    "[不符合]"))
assert rt, "未下发 remember_token"
exp = re.search(r"Expires=([^;]+)", rt[0])
print("  %s 该 cookie 带过期时间：%s"
      % ("[符合]" if exp else "[不符合]", exp.group(1) if exp
    else "无（会话 cookie）"))
assert exp, "remember_token 无过期时间，关浏览器即失效"
# 3) 不勾选时不应下发 remember_token
c2 = app.test_client()
r2 = c2.post("/login", data={"username": "admin",
    "password": "admin123"},
                follow_redirects=False)
rt2 = [h for h in r2.headers.getlist("Set-Cookie") if
    h.startswith("remember_token=")]
print("  %s 未勾选时不下发 remember_token" % ("[符合]" if not rt2
    else "[不符合]"))
assert not rt2
# 4) 携带 remember_token 的新会话应自动登录
c3 = app.test_client()
c3.set_cookie("remember_token",
              rt[0].split(";")[0].split("=", 1)[1],
    domain="localhost")
r3 = c3.get("/", follow_redirects=False)
print("  %s 仅凭 remember_token 访问首页可自动登录（状态码 %d）"
      % ("[符合]" if r3.status_code == 200 else "[不符合]",
    r3.status_code))
assert r3.status_code == 200
# ------------------------------------------------------------------
    ----------------
print()
print(LINE)
print("二、删除功能")
print(LINE)
client.post("/login", data={"username": "admin",
    "password": "admin123"})
# 项目：列表页须有删除按钮
html = client.get("/project/").get_data(as_text=True)
has_del = "/delete" in html and "btn--danger" in html
print("  %s 项目列表含删除按钮" % ("[符合]" if has_del else "[不符合]"))
```

```
assert has_del, "项目列表缺少删除按钮"
print("  %s 删除前有二次确认" % ("[符合]" if "confirm(" in html else
    "[不符合]"))
assert "confirm(" in html
#  图斑列表：地址用的是项目主键，不是图斑主键
with app.app_context():
    from app.models import MinePatch, Project
    import seed_demo
    patch = seed_demo.demo_patch()
    patch_id = patch.id
    project_id = patch.project_id
html = client.get("/project/%d/patches" %
    project_id).get_data(as_text=True)
print("  %s 图斑列表含删除按钮" % ("[符合]" if "/project/patch/" in
    html and "btn--danger" in html else "[不符合]"))
assert "/project/patch/" in html and "btn--danger" in html
#  微地质单元
html = client.get("/unit/%d" % patch_id).get_data(as_text=Tr
    ue)
print("  %s 微单元列表含删除按钮" % ("[符合]" if "/unit/delete/" in
    html else "[不符合]"))
assert "/unit/delete/" in html
print("  %s 微单元编辑为独立页面（表单不再嵌在表格行内）"
      % ("[符合]" if "/unit/edit/" in html else "[不符合]"))
#  标准
html = client.get("/standard/").get_data(as_text=True)
print("  %s 标准列表含删除按钮" % ("[符合]" if "btn--danger" in html
    else "[不符合]"))
assert "btn--danger" in html
#  实际执行一次删除：新建项目 → 删除 → 确认消失
client.post("/project/add", data={"name": "待删除测试项目",
    "code": "DEL-1"})
with app.app_context():
    from app.models import db, Project
    tmp = Project.query.filter_by(name="待删除测试项目").first()
    tmp_id = tmp.id
html = client.get("/project/").get_data(as_text=True)
print("  %s 新建项目成功" % ("[符合]" if "待删除测试项目" in html else
    "[不符合]"))
r = client.post("/project/%d/delete" % tmp_id,
    follow_redirects=True)
html = r.get_data(as_text=True)
gone = "待删除测试项目" not in html
print("  %s 删除项目后列表中不再出现" % ("[符合]" if gone else "[不符合]"))
assert gone
print("  %s 有删除成功提示" % ("[符合]" if "已删除" in html else
    "[不符合]"))
print()
print(LINE)
print("界面修复自检全部通过 √")
```

```
print(LINE)
# ==== 测试/test_new_features.py ====
# -*- coding: utf-8 -*-
"""补齐模块自检：操作日志与用户维护、遥感影像、指标与多专家矩阵、
批量导入、图表与 PDF 导出。
对应计划书功能清单与系统实现的差异核对结论。
"""
import io
import json
import os
import re
import sys
sys.stdout.reconfigure(encoding="utf-8")
import fitz
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, User, Project, MinePatch,
    MicroUnit, IndicatorData,
                        Indicator, ExpertMatrix,
    OperationLog)
from app.utils.schema import ensure_schema
from app.utils import browser, charts
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    UPLOAD_FOLDER = os.path.join(BASE, "app", "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
app = create_app(TestConfig)
with app.app_context():
    db.create_all()
    ensure_schema(db)
    ExpertMatrix.query.delete()
    db.session.commit()
    # 固定取内置演示案例的图斑，避免受库中用户自建项目 / 验证算例的影响
    import seed_demo
    patch_id = seed_demo.demo_patch().id
```

```
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
def ok(flag, label, extra=""):
    print("  %s %-44s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    assert flag, label
# ------------------------------------------------------------
    ----------------
print(LINE)
print("操作日志与用户维护")
print(LINE)
before = None
with app.app_context():
    before = OperationLog.query.count()
c.post("/project/add", data={"name": "日志核对项目", "code":
    "LOG-T"})
with app.app_context():
    after = OperationLog.query.count()
    last = OperationLog.query.order_by(OperationLog.id.desc(
    )).first()
ok(after == before + 1, "写操作被写入操作日志", "%d -> %d" % (before,
    after))
ok(last.action == "新建项目", "日志记录了可读动作名", last.action)
ok("project" in (last.target_type or ""), "日志记录了对象类型",
    last.target_type)
r = c.get("/profile?tab=logs")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "操作日志" in body, "操作日志分类可访问")
ok("新建项目" in body, "日志页显示操作记录")
# 账号编辑 / 删除（职务只有管理员与操作员两种）
with app.app_context():
    u = User(username="tmp_user", role="operator",
             password_hash=User.query.first().password_hash)
    db.session.add(u); db.session.commit()
    uid = u.id
r = c.get("/users/%d/edit" % uid)
ok(r.status_code == 200, "账号编辑页可访问")
# 保护校验要趁系统里只有一名管理员时做：
# 先降级唯一的 admin 应被拒绝，再把临时账号提为管理员看修改是否生效
with app.app_context():
    admin_id = User.query.filter_by(role="admin").first().id
c.post("/users/%d/edit" % admin_id,
       data={"role": "operator", "active": "1"})
with app.app_context():
    ok(db.session.get(User, admin_id).role == "admin",
    "拒绝把最后一个管理员降级")
c.post("/users/%d/edit" % uid, data={"role": "admin",
    "active": "1"})
with app.app_context():
```

```
    ok(db.session.get(User, uid).role == "admin",
    "职务已修改并保存")
c.post("/users/%d/delete" % uid, follow_redirects=True)
with app.app_context():
    ok(db.session.get(User, uid) is None, "账号删除生效")
# 清理日志核对用的项目
with app.app_context():
    for p in Project.query.filter_by(name="日志核对项目").all():
        db.session.delete(p)
    db.session.commit()
# ----------------------------------------------------------------
    ----------------
print()
print(LINE)
print("遥感影像挂接")
print(LINE)
doc = fitz.open()
pg = doc.new_page(width=320, height=200)
pg.insert_text((20, 60), "RS image", fontname="china-s",
    fontsize=16)
png = doc[0].get_pixmap(dpi=100).tobytes("png")
doc.close()
r = c.post("/project/patch/%d/image" % patch_id,
            data={"image": (io.BytesIO(png), "影像.png"),
                  "image_note": "2025年6月无人机正射影像"},
            content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("已挂接" in body and "<img" in body, "影像上传并显示")
with app.app_context():
    rel = db.session.get(MinePatch, patch_id).image_path
ok(bool(rel), "数据库记录影像路径", rel)
abs_path = os.path.join(TestConfig.UPLOAD_FOLDER,
    rel.replace("uploads/", "", 1))
ok(os.path.isfile(abs_path), "影像文件落盘")
ok(c.get("/static/" + rel).status_code == 200, "影像可经 URL
    访问")
r = c.post("/project/patch/%d/image" % patch_id,
            data={"image": (io.BytesIO(b"x"), "bad.txt")},
            content_type="multipart/form-data",
    follow_redirects=True)
ok("不支持" in r.get_data(as_text=True), "拒绝非图片格式")
c.post("/project/patch/%d/image/delete" % patch_id,
    follow_redirects=True)
with app.app_context():
    ok(db.session.get(MinePatch, patch_id).image_path is
    None, "影像移除后字段清空")
ok(not os.path.isfile(abs_path), "影像文件已删除")
# ----------------------------------------------------------------
    ----------------
```

```
print()
print(LINE)
print("指标定义维护与多专家矩阵")
print(LINE)
r = c.get("/ahp/indicators")
body = r.get_data(as_text=True)
with app.app_context():
    n_ind = Indicator.query.count()
ok(r.status_code == 200 and n_ind > 0, "指标定义已入库并可查看", "%d
    项" % n_ind)
ok(all(k in body for k in ("一级指标", "二级指标", "三级指标")),
   "指标树按层级展示")
ok("岩质边坡" in body and "平台" in body, "显示各指标启用的微地质单元")
c.post("/ahp/indicators/coverage/edit",
       data={"name": "植被覆盖度", "unit": "%", "ideal_type":
    "benefit",
             "ideal_value": "75", "enabled_units": ["岩质边坡"],
             "description": "自检修改"})
with app.app_context():
    ok(Indicator.query.filter_by(code="coverage").first().de
    scription == "自检修改",
       "指标定义可编辑")
c.post("/ahp/indicators/reset")
with app.app_context():
    ok("自检修改" not in (Indicator.query.filter_by(code="covera
    ge").first().description or ""),
       "可按论文取值重置")
base = {"m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24"}
for name in ("张工", "李工", "王工"):
    d = dict(base); d.update({"action": "add", "unit_type":
    "碎石边坡",
                              "level": "1", "expert_name":
    name})
    c.post("/ahp/experts", data=d)
with app.app_context():
    ok(ExpertMatrix.query.filter_by(unit_type="碎石边坡",
    level=1).count() == 3,
       "录入 3 位专家的判断矩阵")
r = c.get("/ahp/experts/aggregate?unit_type=碎石边坡&level=1")
body = r.get_data(as_text=True)
ok("汇总后权重" in body and "各专家矩阵的一致性" in body,
   "多专家矩阵几何平均汇总可用")
ws = re.search(r"<th scope=\"row\">权重</th>\s*(.*?)</tr>",
    body, re.S)
vals = re.findall(r">([0-9.]+)</td>", ws.group(1)) if ws
    else []
ok(len(vals) == 3 and abs(sum(map(float, vals)) - 1.0) <
    0.01,
   "汇总权重归一", "、".join(vals))
ok(abs(float(vals[0]) - 0.44) < 0.02, "汇总权重接近论文表 3 取值",
```

```
        vals[0] if vals else "")                                                11929
# ------------------------------------------------------------------------11930
        ------------------                                                      11931
print()                                                                         11932
print(LINE)                                                                     11933
print("批量导入")                                                               11934
print(LINE)                                                                     11935
r = c.get("/indicator/import/template")                                         11936
tpl = r.data.decode("utf-8-sig")                                                11937
ok(r.status_code == 200 and "单元类型" in tpl, "导入模板可下载")                11938
ok("土石比" in tpl and "植被覆盖度" in tpl, "模板包含指标列")                   11939
csv_text = (                                                                    11940
        "单元类型,土石比,土层厚度,pH,有机质,Hg,As,Ni,Cu,Cd,Pb,植被覆盖度,Shannon,坡度11941
        ,坡向\n"                                                                11942
        "碎石边坡,0.34,10,9.24,4.89,0.05,5.17,12.60,11.70,0.04,6.50,11943
        87.92,0.60,28.75,阳坡\n"                                                11944
        "平台,0.56,30,8.72,10.92,0.02,3.13,12.33,11.14,0.09,9.98,51945
        8.42,0.72,,,\n"                                                         11946
        "不存在单元,1,1,7,1,,,,,,,,,,\n"                                       11947
)                                                                               11948
r = c.post("/indicator/import",                                                 11949
                data={"patch_id": str(patch_id),                                11950
                      "datafile": (io.BytesIO(csv_text.encode("ut11951
        f-8")), "d.csv")},                                                      11952
                content_type="multipart/form-data",                             11953
        follow_redirects=True)                                                  11954
body = r.get_data(as_text=True)                                                 11955
ok("导入结果" in body, "导入结果页展示")                                        11956
ok("不存在单元" in body, "错误行被报告")                                        11957
with app.app_context():                                                         11958
        u = MicroUnit.query.filter_by(patch_id=patch_id,                        11959
        unit_type="碎石边坡").first()                                           11960
        ind = IndicatorData.query.filter_by(unit_id=u.id).first(11961
        )                                                                       11962
        ok(abs(ind.soil_ratio - 0.34) < 1e-6 and                                11963
        abs(ind.coverage - 87.92) < 1e-6,                                       11964
            "导入数据已写入数据库")                                             11965
r = c.post("/indicator/import",                                                 11966
                data={"patch_id": str(patch_id),                                11967
                      "datafile": (io.BytesIO(b"x"), "d.pdf")},                 11968
                content_type="multipart/form-data",                             11969
        follow_redirects=True)                                                  11970
ok("仅支持" in r.get_data(as_text=True), "拒绝不支持的导入格式")                11971
# ------------------------------------------------------------------------11972
        ------------------                                                      11973
print()                                                                         11974
print(LINE)                                                                     11975
print("图表与 PDF 导出")                                                       11976
print(LINE)                                                                     11977
svg = charts.hbars([("岩质边坡", 0.54, "0.5401"), ("平台", 0.84,               11978
```

```
        "0.8376")])                                                    11979
ok("<svg" in svg and "0.8376" in svg, "条形图生成且带数值标签")          11980
svg2 = charts.donut([("植被指标", 0.6, "0.6000"), ("土壤指标", 0.4,    11981
        "0.4000")])                                                    11982
ok("<svg" in svg2 and "占比" in svg2, "环形图生成且带图例")              11983
#  报告是评价结论的呈现，需先真实执行一次评价才能打开                  11984
with app.app_context():                                                11985
    from app.views.evaluate import load_patch_input,                   11986
        save_evaluation                                                11987
    save_evaluation(patch_id, load_patch_input(patch_id))              11988
r = c.get("/report/%d" % patch_id)                                     11989
body = r.get_data(as_text=True)                                        11990
ok(body.count("<svg") >= 2, "得分明细页含图表", "%d 个 SVG" %          11991
    body.count("<svg"))                                                11992
exe = browser.browser_path()                                           11993
if exe is None:                                                        11994
    print("  [跳过] 未找到浏览器，PDF 导出无法验证")                   11995
else:                                                                  11996
    r = c.get("/report/%d/pdf" % patch_id)                             11997
    data = r.data                                                      11998
    ok(data[:4] == b"%PDF", "PDF 报告导出成功", "%d 字节" %            11999
    len(data))                                                         12000
    d = fitz.open(stream=data, filetype="pdf")                         12001
    txt = "".join(d[i].get_text() for i in                             12002
    range(d.page_count))                                               12003
    ok("自然恢复适宜性评价报告" in txt, "PDF 含报告标题")              12004
    ok("评价结论" in txt and "得分明细" in txt, "PDF 含结论与明细章节") 12005
    ok(d.page_count >= 2, "PDF 页数合理", "%d 页" % d.page_count)      12006
    d.close()                                                          12007
print()                                                                12008
print(LINE)                                                            12009
print("补齐模块自检全部通过 √")                                       12010
print(LINE)                                                            12011
#  ==== 测试/test_workflow.py ====                                     12012
#  -*- coding: utf-8 -*-                                               12013
"""流程门禁自检：前置步骤未完成时，后续模块必须拦得住。                12014
论文 4.1 节确定的评价流程存在先后关系，缺任何一环都不应产出评价结论。 12015
本测试逐个构造「缺前置」的场景，验证系统不会放行。                    12016
"""                                                                    12017
import os                                                              12018
import sys                                                             12019
sys.stdout.reconfigure(encoding="utf-8")                               12020
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，           12021
#  使 `from app import ...` 与 app/app.db 的定位不受影响。             12022
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil           12023
    e__)))                                                             12024
if ROOT not in sys.path:                                               12025
    sys.path.insert(0, ROOT)                                           12026
from app import create_app                                             12027
from app.models import (db, Project, MinePatch, MicroUnit,             12028
```

```
    IndicatorData,                                                  12029
                            ConstraintCheck, EvaluationResult)      12030
from app.services import workflow as wf                             12031
LINE = "=" * 72                                                     12032
BASE = ROOT                                                         12033
class TestConfig:                                                   12034
    SECRET_KEY = "test"                                             12035
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +                        12036
    os.path.join(BASE, "app", "app.db")                             12037
    SQLALCHEMY_TRACK_MODIFICATIONS = False                          12038
    WTF_CSRF_ENABLED = False                                        12039
    TESTING = True                                                  12040
app = create_app(TestConfig)                                        12041
c = app.test_client()                                               12042
c.post("/login", data={"username": "admin", "password":             12043
    "admin123"})                                                    12044
_fails = []                                                         12045
def ok(flag, label, extra=""):                                      12046
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",       12047
    label, extra))                                                  12048
    if not flag:                                                    12049
        _fails.append(label)                                        12050
def new_patch(tag):                                                 12051
    with app.app_context():                                         12052
        p = Project(name="门禁-" + tag, code="WF-" + tag)           12053
        db.session.add(p); db.session.flush()                       12054
        patch = MinePatch(project_id=p.id, name="门禁图斑-" +       12055
    tag)                                                            12056
        db.session.add(patch); db.session.flush()                   12057
        pid = patch.id                                              12058
        db.session.commit()                                         12059
    return pid                                                      12060
def add_units(pid):                                                 12061
    with app.app_context():                                         12062
        db.session.add(MicroUnit(patch_id=pid,                      12063
    unit_type="碎石边坡",                                           12064
                                     area=1.0, ratio=1.0))          12065
        db.session.commit()                                         12066
def add_constraint(pid, passed=True):                               12067
    with app.app_context():                                         12068
        db.session.add(ConstraintCheck(                             12069
            patch_id=pid, policy_pass=True,                         12070
    location_pass=passed,                                           12071
            pollution_pass=True, geo_pass=True,                     12072
    farmland_pass=True,                                             12073
            passed=passed, reason="全部通过" if passed else         12074
    "未通过：区位约束"))                                            12075
        db.session.commit()                                         12076
def add_data(pid):                                                  12077
    with app.app_context():                                         12078
```

```
            u = MicroUnit.query.filter_by(patch_id=pid).first()
            db.session.add(IndicatorData(unit_id=u.id,
        soil_ratio=0.34, thickness=10,
                                         ph=9.24,
        organic_matter=4.89, coverage=87.92,
                                         shannon=0.60,
        slope=28.75, aspect="阳坡"))
            db.session.commit()
def cleanup(pid, tag):
        with app.app_context():
            from app.models import ScoreDetail
            # 先删得分明细，再删评价记录，避免留下孤儿明细
            rids = [r.id for r in EvaluationResult.query.filter_
        by(patch_id=pid)]
            if rids:
                ScoreDetail.query.filter(ScoreDetail.result_id.i
        n_(rids)).delete(
                    synchronize_session=False)
            EvaluationResult.query.filter_by(patch_id=pid).delet
        e()
            ConstraintCheck.query.filter_by(patch_id=pid).delete
        ()
            uids = [u.id for u in MicroUnit.query.filter_by(pat
        h_id=pid)]
            if uids:
                IndicatorData.query.filter(IndicatorData.unit_id
        .in_(uids)).delete(
                    synchronize_session=False)
            MicroUnit.query.filter_by(patch_id=pid).delete()
            m = db.session.get(MinePatch, pid)
            if m:
                db.session.delete(m)
            pr = Project.query.filter_by(name="门禁-" +
        tag).first()
            if pr:
                db.session.delete(pr)
            db.session.commit()
def results_of(pid):
        with app.app_context():
            return EvaluationResult.query.filter_by(patch_id=pid
        ).count()
def post_eval(pid):
        return c.post("/evaluate/%d" % pid, data={},
        follow_redirects=False)
def get_report(pid):
        return c.get("/report/%d" % pid, follow_redirects=False)
# ------------------------------------------------------------------
        ------------------
print(LINE)
print("一、前置条件缺失时，评价必须被拦住")
```

```
print(LINE)
print()
print("场景　1：连微地质单元都没划分")
pid = new_patch("1")
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未划分单元 → 不产生评价结果",
   "HTTP %d，结果数 %d" % (r.status_code, n))
with app.app_context():
    pr = wf.patch_progress(pid)
steps = [b["step"] for b in pr["blocking"]]
ok(not pr["ready"] and "units" in steps,
   "拦原因包含「划分微地质单元」", "、".join(steps))
ok(steps == ["constraint", "units"],
   "拦截顺序符合论文　4．1　节：限制性判定在前，划分单元在后",
   "、".join(steps))
cleanup(pid, "1")
print()
print("场景　2：有单元，但未做限制性条件判定")
pid = new_patch("2"); add_units(pid); add_data(pid)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未做限制性判定 → 不产生评价结果",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "constraint" for b in pr["blocking"]),
   "拦原因指向「执行限制性条件判定」")
cleanup(pid, "2")
print()
print("场景　3：限制性条件判定【未通过】")
pid = new_patch("3"); add_units(pid); add_data(pid);
    add_constraint(pid, passed=False)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "判定未通过 → 不进入定量评价",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(not pr["constraint"]["passed"], "流程状态显示判定未通过")
blocked = [b for b in pr["blocking"] if b["step"] ==
    "constraint"]
# 提示只写「缺什么」：label 说明未通过，reason 直接引用判定记录里没过的项
ok(bool(blocked) and "未通过" in blocked[0]["label"],
   "拦提示指明限制性条件判定未通过", blocked[0]["label"] if blocked else
    "")
ok(bool(blocked) and "区位约束" in (blocked[0]["reason"] or ""),
   "并把具体没通过的条件带出来", blocked[0]["reason"] if blocked else "")
cleanup(pid, "3")
print()
```

```
print("场景 4：判定通过，但指标数据全空")
pid = new_patch("4"); add_units(pid); add_constraint(pid,
    passed=True)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "数据未录入 → 不产生评价结果",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "data" for b in pr["blocking"]),
   "拦原因指向「录入指标数据」")
ok(pr["data"]["missing"] == ["碎石边坡"],
   "指出具体缺哪个单元的数据", "、".join(pr["data"]["missing"]))
cleanup(pid, "4")
print()
print("场景 5：单元面积占比合计不为 100%")
pid = new_patch("5"); add_constraint(pid, passed=True)
with app.app_context():
    u = MicroUnit(patch_id=pid, unit_type="碎石边坡", area=1.0,
    ratio=0.5)
    db.session.add(u); db.session.commit()
    db.session.add(IndicatorData(unit_id=u.id,
    soil_ratio=0.3, thickness=10,
                                 ph=8.0,
    organic_matter=5.0, coverage=80.0,
                                 shannon=0.6, slope=20.0,
    aspect="阳坡"))
    db.session.commit()
r = post_eval(pid)
ok(results_of(pid) == 0, "占比不足 100% → 不产生评价结果")
cleanup(pid, "5")
print()
print(LINE)
print("二、前置条件齐备时才放行")
print(LINE)
pid = new_patch("OK"); add_units(pid); add_constraint(pid,
    passed=True); add_data(pid)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(pr["ready"], "单元 + 判定通过 + 数据齐备 → 判定为可执行")
r = post_eval(pid)
n = results_of(pid)
ok(n == 1, "此时执行评价成功产生结果", "结果数 %d" % n)
print("  重定向到：", r.headers.get("Location"))
print()
print(LINE)
print("三、报告不得先于评价产生")
print(LINE)
pid2 = new_patch("R")    # 什么都不做
r = get_report(pid2)
```

```
ok(r.status_code == 302 and "/report/" ==
      (r.headers.get("Location") or ""),
    "前置缺失时 /report/<id> 被退回列表", "HTTP %d" % r.status_code)
for suffix in ("/pdf", "/word"):
      rr = c.get("/report/%d%s" % (pid2, suffix),
      follow_redirects=False)
      ok(rr.status_code == 302, "导出 %s 同样被拦住" % suffix, "HTTP
      %d" % rr.status_code)
cleanup(pid2, "R")
cleanup(pid, "OK")
print()
print(LINE)
print("四、前置齐备但从未执行评价时，报告仍不得产生")
print(LINE)
#  报告是评价结论的呈现：即使前置步骤都做完了，只要没真正执行过定量评价，
#  也不应凭空推算出结论。
pid3 = new_patch("N")
add_units(pid3); add_constraint(pid3, passed=True);
      add_data(pid3)
with app.app_context():
      pr = wf.report_progress(pid3)
ok(not pr["ready"], "无评价记录 → 报告流程判定为未就绪")
ok(bool(pr["blocking"]) and pr["blocking"][0]["step"] ==
      "evaluate",
    "拦原因指向「执行适宜性评价」")
r = get_report(pid3)
ok(r.status_code == 302 and results_of(pid3) == 0,
    "直接访问 /report/<id> 被退回，且不产生任何评价结果",
    "HTTP %d" % r.status_code)
#  真正执行评价之后，报告才放行，且取值与评价记录一致
r = post_eval(pid3)
ok(results_of(pid3) == 1, "先执行评价 → 产生评价记录")
with app.app_context():
      pr = wf.report_progress(pid3)
      rec = pr["result"]
ok(pr["ready"] and rec is not None, "有评价记录后，报告流程就绪")
rr = get_report(pid3)
body = rr.get_data(as_text=True)
ok(rr.status_code == 200, "此时可打开得分明细页", "HTTP %d" %
      rr.status_code)
ok(("%.4f" % rec.final_score) in body,
    "报告数值取自评价记录，而非重新计算",
    "F=%.4f" % rec.final_score)
cleanup(pid3, "N")
print()
print(LINE)
print("五、内置演示数据的清理不得波及用户自建数据")
print(LINE)
#  回归保护：seed_demo.clean_case() 曾执行 Project.query.delete(),
#  导致用户新建的项目在每次启动时被连带删除。
```

```
import seed_demo as sd
with app.app_context():
    keep_p = Project(name="用户自建项目-守护", code="KEEP-001")
    db.session.add(keep_p); db.session.flush()
    keep_patch = MinePatch(project_id=keep_p.id,
    name="用户自建图斑-守护")
    db.session.add(keep_patch); db.session.flush()
    keep_pid = keep_p.id
    keep_patch_id = keep_patch.id
    db.session.commit()
with app.app_context():
    sd.clean_case(reset=False)
with app.app_context():
    still_p = db.session.get(Project, keep_pid)
    still_patch = db.session.get(MinePatch, keep_patch_id)
ok(still_p is not None, "清理演示数据后，用户自建项目仍存在")
ok(still_patch is not None, "清理演示数据后，用户自建图斑仍存在")
#  演示数据清掉后应能重新写入，保证演示环境可恢复
sd.seed()
with app.app_context():
    demo_back = Project.query.filter_by(code=sd.DEMO_CODE).f
    irst()
ok(demo_back is not None, "演示案例可重新写入（幂等恢复）")
with app.app_context():
    m = db.session.get(MinePatch, keep_patch_id)
    if m:
        db.session.delete(m)
    p = db.session.get(Project, keep_pid)
    if p:
        db.session.delete(p)
    db.session.commit()
print()
print(LINE)
print("六、同一图斑不得重复划分同类型微地质单元")
print(LINE)
#  论文 4.1 节把图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
#  允许重复会让面积占比、按类型组织的指标数据与批量导入对不上号。
pid4 = new_patch("D")
c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "1.0"},
        follow_redirects=True)
with app.app_context():
    n1 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n1 == 1, "首次添加「平台」成功", "单元数 %d" % n1)
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "2.0"},
            follow_redirects=True)
with app.app_context():
    n2 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n2 == 1, "重复添加同类型被拒绝", "单元数 %d" % n2)
```

```
ok("同一类型只能划分一个" in r.get_data(as_text=True), "给出重复提示")
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "火山口",
    "area": "1.0"},
            follow_redirects=True)
with app.app_context():
    n3 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n3 == 1, "非法单元类型被拒绝", "单元数 %d" % n3)
c.post("/unit/%d/add" % pid4, data={"unit_type": "碎石边坡",
    "area": "3.0"},
        follow_redirects=True)
with app.app_context():
    uid = MicroUnit.query.filter_by(patch_id=pid4,
                                    unit_type="碎石边坡").first(
    ).id
r = c.post("/unit/edit/%d" % uid,
            data={"unit_type": "平台", "area": "3.0"},
    follow_redirects=True)
with app.app_context():
    types = sorted(u.unit_type for u in
                   MicroUnit.query.filter_by(patch_id=pid4)
    all())
ok(types == ["平台", "碎石边坡"], "改类型时同样拦截重复", "、".join(types))
cleanup(pid4, "D")
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("流程门禁自检全部通过 √")
print(LINE)
#  ==== 测试/test_weights.py ====
#  -*- coding: utf-8 -*-
"""权重生效链路自检：AHP 算出的权重必须真正参与评价计算。
此前 weight_schemes / weight_items 两张表从未被写入，评价引擎也只用
代码里的论文表 3 默认权重，相当于权重模块算完即弃。本测试守住这条链路：
    默认 → 论文表 3
    保存 AHP 权重 → 评价改用新权重，结果随之变化
    恢复默认 → 结果回到论文取值
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file
    __)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
```

```
from app import create_app
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult,
    WeightScheme,
                        WeightItem)
from app.services import workflow as wf
from app.services import weight_service as ws
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
TAG = "W"
UNIT_TYPE = "碎石边坡"
def build_case():
    """建一个只含单一微地质单元的图斑，便于观察权重变化对得分的影响。"""
    with app.app_context():
        p = Project(name="权重测试-%s" % TAG, code="WEIGHT-%s"
    % TAG)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="权重测试图斑-%s"
    % TAG)
        db.session.add(patch); db.session.flush()
        db.session.add(MicroUnit(patch_id=patch.id,
    unit_type=UNIT_TYPE,
                                 area=1.0, ratio=1.0))
        db.session.flush()
        u = MicroUnit.query.filter_by(patch_id=patch.id).fi
    st()
        db.session.add(IndicatorData(
            unit_id=u.id, soil_ratio=0.34, thickness=10,
    ph=9.24,
            organic_matter=4.89, coverage=87.92,
    shannon=0.60,
            slope=28.75, aspect="阳坡"))
```

```
        db.session.add(ConstraintCheck(
            patch_id=patch.id, policy_pass=True,
    location_pass=True,
            pollution_pass=True, geo_pass=True,
    farmland_pass=True,
            passed=True, reason="全部通过"))
        db.session.commit()
        return patch.id
def cleanup(pid):
    with app.app_context():
        from app.models import ScoreDetail
        rids = [r.id for r in EvaluationResult.query.filter_
    by(patch_id=pid)]
        if rids:
            ScoreDetail.query.filter(ScoreDetail.result_id.i
    n_(rids)).delete(
                synchronize_session=False)
        EvaluationResult.query.filter_by(patch_id=pid).delet
    e()
        ConstraintCheck.query.filter_by(patch_id=pid).delete
    ()
        uids = [u.id for u in MicroUnit.query.filter_by(pat
    h_id=pid)]
        if uids:
            IndicatorData.query.filter(IndicatorData.unit_id
    .in_(uids)).delete(
                synchronize_session=False)
        MicroUnit.query.filter_by(patch_id=pid).delete()
        m = db.session.get(MinePatch, pid)
        if m:
            db.session.delete(m)
        p = Project.query.filter_by(code="WEIGHT-%s" %
    TAG).first()
        if p:
            db.session.delete(p)
        db.session.commit()
def run_eval(pid):
    c.post("/evaluate/%d" % pid, data={},
    follow_redirects=False)
    with app.app_context():
        return EvaluationResult.query.filter_by(patch_id=pid
    ) \
            .order_by(EvaluationResult.id.desc()).first().fi
    nal_score
print(LINE)
print("一、默认应使用论文表 3 的权重")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
    w, label, scheme = ws.effective_weights(UNIT_TYPE)
```

```
ok(scheme is None and "论文表" in label, "未配置时权重来源为论文表 3",
   label)
ok(abs(w["l1"]["土壤指标"] - 0.44) < 1e-9, "一级权重取自论文表 3",
   "土壤指标 %.2f" % w["l1"]["土壤指标"])
pid = build_case()
base_score = run_eval(pid)
print("  默认权重下的评价值 F = %.4f" % base_score)
print()
print(LINE)
print("二、保存 AHP 算出的权重后，评价应改用新权重")
print(LINE)
with app.app_context():
    # 模拟 AHP 计算结果：把土壤指标权重提高、植被与地形降低（仍和为 1）
    ws.save_weights(UNIT_TYPE, [("soil", 0.70), ("veg",
    0.20),
                                ("terrain", 0.10)])
    n_items = WeightItem.query.count()
ok(n_items >= 3, "weight_items 表已写入权重条目", "%d 条" % n_items)
with app.app_context():
    w2, label2, scheme2 = ws.effective_weights(UNIT_TYPE)
ok(scheme2 is not None and "自定义" in label2, "权重来源变为自定义方案",
   label2)
ok(abs(w2["l1"]["土壤指标"] - 0.70) < 1e-9, "新权重已生效",
   "土壤指标 %.2f" % w2["l1"]["土壤指标"])
ok(abs(w2["l1"]["植被指标"] - 0.20) < 1e-9, "同批保存的其他项也已生效")
ok(abs(w2["l2"]["土壤指标"]["土石比"] - 0.25) < 1e-9,
   "未覆盖的二级权重仍保留论文取值",
   "土石比 %.2f" % w2["l2"]["土壤指标"]["土石比"])
new_score = run_eval(pid)
print("  新权重下的评价值 F = %.4f" % new_score)
ok(abs(new_score - base_score) > 1e-4,
   "权重变化确实改变了评价结果",
   "%.4f → %.4f" % (base_score, new_score))
print()
print(LINE)
print("三、结果应记录所用权重来源")
print(LINE)
import json
with app.app_context():
    rec = EvaluationResult.query.filter_by(patch_id=pid) \
          .order_by(EvaluationResult.id.desc()).first()
    stored = json.loads(rec.unit_scores)
ok(any(u.get("weight_source") for u in stored),
   "评价记录中标注了权重来源",
   stored[0].get("weight_source") or "")
print()
print(LINE)
print("四、恢复默认后应回到论文取值")
print(LINE)
with app.app_context():
```

```
    ws.reset_weights(UNIT_TYPE)                                          12529
    w3, label3, scheme3 = ws.effective_weights(UNIT_TYPE)                12530
ok(scheme3 is None and "论文表" in label3, "已恢复为论文表 3 权重",          12531
    label3)                                                              12532
back_score = run_eval(pid)                                               12533
print("  恢复后评价值 F = %.4f（默认时 %.4f）" % (back_score,               12534
    base_score))                                                         12535
ok(abs(back_score - base_score) < 1e-9, "评价结果与默认时一致")            12536
print()                                                                  12537
print(LINE)                                                              12538
print("五、权重首页概览")                                                 12539
print(LINE)                                                              12540
with app.app_context():                                                  12541
    rows = ws.summary()                                                  12542
ok(len(rows) == 3, "三种微地质单元均有权重来源说明", "%d 行" % len(rows))  12543
r = c.get("/ahp/")                                                       12544
body = r.get_data(as_text=True)                                          12545
ok(r.status_code == 200 and "当前生效权重" in body, "权重首页正常展示")    12546
print()                                                                  12547
print(LINE)                                                              12548
print("六、一致性不通过的矩阵不得保存生效")                               12549
print(LINE)                                                              12550
with app.app_context():                                                  12551
    ws.reset_weights(UNIT_TYPE)                                          12552
n = 3                                                                    12553
# 明显矛盾的矩阵：a12=9, a13=1/9, a23=9 —— CR 必然超限                    12554
mats = [(0, 1, "9"), (0, 2, "0.1111"), (1, 2, "9")]                      12555
data = {"csrf_token": "", "action": "save"}                              12556
for i, j, v in mats:                                                     12557
    data["m_%d_%d" % (i, j)] = v                                         12558
r = c.post("/ahp/matrix?unit_type=%s&level=1" % UNIT_TYPE,               12559
    data=data,                                                           12560
               follow_redirects=True)                                    12561
with app.app_context():                                                  12562
    _, label4, scheme4 = ws.effective_weights(UNIT_TYPE)                 12563
ok(scheme4 is None, "CR 超限的矩阵未被保存生效", label4)                  12564
ok("一致性" in r.get_data(as_text=True), "页面给出了未通过一致性的提示")    12565
cleanup(pid)                                                             12566
print()                                                                  12567
print(LINE)                                                              12568
if _fails:                                                               12569
    print("存在未通过项：")                                               12570
    for f in _fails:                                                     12571
        print("  -", f)                                                 12572
    sys.exit(1)                                                          12573
print("权重生效链路自检全部通过 √")                                       12574
print(LINE)                                                              12575
# ==== app/services/__init__.py ====                                     12576
# -*- coding: utf-8 -*-                                                  12577
# ==== app/services/cleanup.py ====                                      12578
```

```
# -*- coding: utf-8 -*-
"""级联清理：删图斑、删项目时，把挂在它下面的数据一并清掉。
为什么单独抽出来：`Project.patches` 与 `MinePatch.units` 上有 ORM 级联，
但图斑下面的**指标数据、限制性判定、评价结果、得分明细、参照生态系统**
都没有级联关系。原先只有「删除图斑」那条路径里手写了清理，
于是「删除项目」会留下一批孤儿行——自检实测：删一个项目后，
上面五张表各残留 1 行（项目与图斑本身都删掉了，数据却还在库里）。
两条删除路径都调用这里，才不会漏。
"""
from app.models import (db, MinePatch, MicroUnit,
    IndicatorData,
                        EvaluationResult, ScoreDetail,
    ConstraintCheck,
                        ReferenceEcosystem)
def delete_patch_children(patch_id):
    """清掉挂在某个图斑下的全部附属数据（不含图斑本身）。
    返回各表清理条数，便于调用方打印或断言。
    """
    counts = {}
    unit_ids = [u.id for u in MicroUnit.query.filter_by(patch_id=patch_id).all()]
    if unit_ids:
        counts["indicator_data"] = IndicatorData.query.filter(
            IndicatorData.unit_id.in_(unit_ids)).delete(synchronize_session=False)
    result_ids = [r.id for r in
                  EvaluationResult.query.filter_by(patch_id=patch_id).all()]
    if result_ids:
        counts["score_details"] = ScoreDetail.query.filter(
            ScoreDetail.result_id.in_(result_ids)).delete(synchronize_session=False)
    counts["evaluation_results"] = EvaluationResult.query.filter_by(
        patch_id=patch_id).delete(synchronize_session=False)
    counts["constraint_checks"] = ConstraintCheck.query.filter_by(
        patch_id=patch_id).delete(synchronize_session=False)
    counts["micro_units"] = MicroUnit.query.filter_by(
        patch_id=patch_id).delete(synchronize_session=False)
    counts["reference_ecosystems"] = ReferenceEcosystem.query.filter_by(
        patch_id=patch_id).delete(synchronize_session=False)
    return counts
def delete_patch(patch_id):
    """删除一个图斑及其全部附属数据。图斑不存在时返回 (None, {})。"""
    patch = db.session.get(MinePatch, patch_id)
    if patch is None:
        return None, {}
```

```
    counts = delete_patch_children(patch_id)
    db.session.delete(patch)
    db.session.commit()
    return patch, counts
def delete_project(project):
    """删除一个项目：先清掉每个图斑的附属数据，再删项目本身。
    项目下的图斑由 `Project.patches` 的 ORM 级联删除，但图斑下面那五张表
    必须先手工清，否则会留孤儿。
    """
    cleaned = 0
    for patch in project.patches.all():
        counts = delete_patch_children(patch.id)
        cleaned += sum(counts.values())
    name = project.name
    db.session.delete(project)
    db.session.commit()
    return name, cleaned
# ==== app/services/unit_config.py ====
# -*- coding: utf-8 -*-
"""标准化参数与必填项：让界面上能改的东西真正参与计算。
以前有个说不通的地方：图斑的「参照生态系统基准」、指标定义里的「理想取值」
和「启用单元」在界面上都能改、也能落库，**但计算链只读代码常量**
（`app/utils/constants.py`），从不查这两张表。于是报告里写着「基准 60%」，
实际按常量 75% 算，同一份报告两个口径。
现在统一从这里取：
- 基准值（植被覆盖度、Shannon）来自图斑的 `reference_ecosystems`；
- 各指标的理想取值来自 `indicators.ideal_value`（能解析成数字的才覆盖，
  像「5~9」「25° 分界」这种区间 / 规则型仍按代码里的算法处理）；
- 启用单元来自 `indicators.enabled_units`（岩质边坡无土壤、平台无地形）。
库里没有行、或值不是数字时一律回落到代码常量 —— 所以历史数据与
`验证数据/` 里的 8 个算例结果不会因为这次改动而变化。
"""
from app.models import Indicator, ReferenceEcosystem
from app.utils.constants import (
    UNIT_TYPES, IDEAL_SOIL_RATIO, IDEAL_THICKNESS,
    IDEAL_ORGANIC,
    IDEAL_COVERAGE, IDEAL_SHANNON, HEAVY_METAL_LIMIT,
    HEAVY_METALS,
)
# 二级指标编码 → 标准化参数名（同时也是 IndicatorData 的字段名）
NUMERIC_IDEALS = {
    "soil_ratio": ("soil_ratio", IDEAL_SOIL_RATIO),
    "thickness": ("thickness", IDEAL_THICKNESS),
    "organic_matter": ("organic_matter", IDEAL_ORGANIC),
    "coverage": ("coverage", IDEAL_COVERAGE),
    "shannon": ("shannon", IDEAL_SHANNON),
}
# 重金属编码 → 金属名
METAL_CODES = {"hm_" + m: m for m in HEAVY_METALS}
# 必填的实测字段（污染指标按论文注记允许「未检测」，故不在必填之列）
```

```
REQUIRED_CODES = ["soil_ratio", "thickness", "ph",
    "organic_matter",
                  "coverage", "shannon", "slope", "aspect"]
# 字段 → 界面上的中文名，用于提示「缺什么」
FIELD_LABELS = {
    "soil_ratio": "土石比", "thickness": "土层厚度", "ph": "pH",
    "organic_matter": "有机质", "coverage": "植被覆盖度",
    "shannon": "生物多样性（Shannon）", "slope": "坡度", "aspect":
    "坡向",
}
def _numeric(text):
    """把理想取值的文本解析成数字；区间、规则型返回 None。"""
    if text is None:
        return None
    try:
        return float(str(text).strip())
    except (TypeError, ValueError):
        return None
class UnitSettings:
    """一个图斑上下文的标准化参数。空对象表示「全部按代码常量」。"""
    def __init__(self, params=None, enabled=None):
        self.params = params or {}              # 覆盖用的标准值
        self.enabled = enabled or {}            # 单元类型 →
    启用的二级指标编码集合
    def ideal(self, code, fallback):
        """取某指标的理想取值：库里有数字就用库里的，否则用常量。"""
        value = self.params.get(code)
        return fallback if value is None else value
    def group_enabled(self, unit_type, l1_name, l1_code):
        """该单元类型下某个一级指标是否启用（库里没配就按代码常量）。"""
        codes = self.enabled.get(unit_type)
        if codes is None:
            return None                         # 交给调用方按常量判断
        return l1_code in codes
    def enabled_codes(self, unit_type):
        """该单元类型启用的全部二级指标编码；库里没配返回 None。"""
        return self.enabled.get(unit_type)
    def required_codes(self, unit_type):
        """该单元类型必填的实测字段。
        按「启用的二级指标」推导：岩质边坡不启用土壤，平台不启用地形。
        库里没配该项时用代码常量里的结构性规则。
        """
        from app.utils.constants import
    UNIT_ENABLED_INDICATORS
        codes = self.enabled.get(unit_type)
        if codes is None:
            groups = UNIT_ENABLED_INDICATORS.get(unit_type,
    ())
            l2_by_group = {
                "土壤指标": ["soil_ratio", "thickness", "ph",
```

```
    "organic_matter"],                                                  12729
                "植被指标": ["coverage", "shannon"],                    12730
                "地形指标": ["slope", "aspect"],                        12731
            }                                                           12732
            codes = {c for g in groups for c in                         12733
    l2_by_group.get(g, [])}                                             12734
        return [c for c in REQUIRED_CODES if c in codes]                12735
    def missing_fields(self, unit_type, data):                          12736
        """返回该单元缺哪些必填项（中文名列表）。"""                    12737
        missing = []                                                    12738
        for code in self.required_codes(unit_type):                     12739
            value = data.get(code)                                      12740
            if value is None or (isinstance(value, str) and             12741
    value.strip() == ""):                                               12742
                missing.append(FIELD_LABELS.get(code, code))            12743
        return missing                                                  12744
    def undetected_metals(self, data):                                  12745
        """按论文注记，未检测的重金属按「未受污染」计入；这里列出是哪些。"""  12746
        measured = data.get("heavy_metals") or {}                       12747
        return [m for m in HEAVY_METALS if measured.get(m)              12748
    is None]                                                            12749
def for_patch(patch_id):                                                12750
    """读某个图斑的标准化参数；库里有什么就用什么，没有就用常量。          12751
    优先级：图斑自己的参照生态系统基准  **高于**  指标定义里的全局理想取值  ——  12752
    前者说的是「这个图斑的参照生态系统是什么样」，后者是「一般情况下的理想值」，  12753
    越具体的越应该说了算。两者都优先于代码常量。                          12754
    """                                                                 12755
    params = {}                                                         12756
    enabled = {}                                                        12757
    for ind in Indicator.query.all():                                   12758
        code = ind.code                                                 12759
        if code in NUMERIC_IDEALS:                                      12760
            value = _numeric(ind.ideal_value)                           12761
            if value is not None:                                       12762
                params[code] = value                                    12763
        elif code in METAL_CODES:                                       12764
            #  重金属的「理想值」存的是限值，作为越小越优的极值使用     12765
            value = _numeric(ind.ideal_value)                           12766
            if value is not None:                                       12767
                params[code] = value                                    12768
        units = (ind.enabled_units or "").strip()                       12769
        if units:                                                       12770
            for unit in UNIT_TYPES:                                     12771
                if unit in units:                                       12772
                    enabled.setdefault(unit,                            12773
    set()).add(code)                                                    12774
    #  再用该图斑的参照生态系统基准覆盖：它是「这个图斑的参照生态系统」，  12775
    #  比指标定义里的全局理想取值更具体，应该它说了算。                  12776
    ref = ReferenceEcosystem.query.filter_by(patch_id=patch_            12777
    id).first()                                                         12778
```

```
    if ref is not None:
        if ref.coverage_benchmark is not None:
            params["coverage"] = float(ref.coverage_benchmark)
    if ref.shannon_benchmark is not None:
        params["shannon"] = float(ref.shannon_benchmark)
    return UnitSettings(params=params, enabled=enabled)
def metal_limits(settings):
    """重金属的极值表：库里有配置的用库里的，其余用常量。"""
    limits = dict(HEAVY_METAL_LIMIT)
    for code, metal in METAL_CODES.items():
        value = settings.params.get(code) if settings else None
        if value is not None:
            limits[metal] = value
    return limits
# ==== app/utils/__init__.py ====
# -*- coding: utf-8 -*-
# ==== app/utils/parsing.py ====
# -*- coding: utf-8 -*-
"""表单数值的安全解析。
背景：页面上的数字输入框原来写成 `float(request.form.get("area"))`，
用户填「约 3.5」或全角数字就会撞上 ValueError —— 页面上是默认英文 500 页。
真人演示时这是最容易踩到的一类。统一走这里：
没填返回 None，填了但不是数字抛 FieldError，让视图 flash 一句中文再回退。
"""
import math
class FieldError(ValueError):
    """表单里某个字段填得不对，携带给用户看的中文提示。"""
def parse_float(raw, field, *, minimum=None, maximum=None):
    """把表单值解析成 float。
    raw       表单原始值（可能是 None / "" / "abc" / "1e400"）
    field     字段名，用于拼提示
    minimum / maximum   可选的取值范围，越界同样抛 FieldError
    """
    if raw is None:
        return None
    text = str(raw).strip()
    if text == "":
        return None
    try:
        value = float(text)
    except (TypeError, ValueError):
        raise FieldError("「%s」请填数字" % field)
    # nan / inf 能通过 float()，但存进库里会把标准化算成满分，必须挡住
    if not math.isfinite(value):
        raise FieldError("「%s」请填正常的数字" % field)
    if minimum is not None and value < minimum:
        raise FieldError("「%s」不能小于 %g" % (field, minimum))
    if maximum is not None and value > maximum:
```

```
            raise FieldError("「%s」不能大于 %g" % (field, maximum))
        return value
def parse_int(raw, field, *, default=None, minimum=None,
        maximum=None):
    """解析整数，用于查询参数与计数类字段。"""
    if raw is None:
        return default
    text = str(raw).strip()
    if text == "":
        return default
    try:
        value = int(text)
    except (TypeError, ValueError):
        raise FieldError("「%s」请填整数" % field)
    if minimum is not None and value < minimum:
        value = minimum
    if maximum is not None and value > maximum:
        value = maximum
    return value
# ==== app/utils/timefmt.py ====
# -*- coding: utf-8 -*-
"""时间显示：库里一律存 UTC，展示时换成服务器本地时间。
为什么不在写库时直接用本地时间：一个系统里时间基准只能有一个。
`app/models.py` 里所有 `created_at` / `checked_at` 都用
    `datetime.utcnow`，
存的是 UTC；如果写库改成本地时间，已经存进库的几千条记录会新旧混着差 8 小时。
所以统一在「展示」这一层换算，老数据不用动。
用法（Jinja）：
    {{ row.created_at | localdt }}                  →
    2026-09-26 23:33
    {{ row.created_at | localdt("%Y-%m-%d %H:%M:%S") }} →
    2026-09-26 23:33:28
"""
from datetime import datetime, timezone
DEFAULT_FORMAT = "%Y-%m-%d %H:%M"
def to_local(value, fmt=DEFAULT_FORMAT, empty="—"):
    """把 UTC 时间换算成服务器本地时间并格式化。
    传进来的值按「无时区即 UTC」处理；带时区的原样换算，便于以后改用
    带时区的时间戳。空值返回占位符，免得页面上出现 None。
    """
    if value is None:
        return empty
    if value.tzinfo is None:
        value = value.replace(tzinfo=timezone.utc)
    return value.astimezone().strftime(fmt)
def now_utc():
    """取当前 UTC 时间（带时区），用于只作展示、不落库的时间戳。"""
    return datetime.now(timezone.utc)
# ==== app/views/__init__.py ====
# -*- coding: utf-8 -*-
```